

nanotag Viewer Program

nanotag

User's Manual

KISSEI COMTEC

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▪ REVISION HISTORY

1 Introduction

Thank you very much for your purchase.
“nanotagViewer” is a research program that supports biological rhythm analysis.

Caution



This software is only available for research purposes.
Not for Medical use.

1.1 Meaning of the mark

Caution



Precautions for use.

Warning



Warning for use.

Prohibition



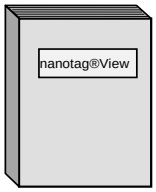
Prohibitions

1.2 Packing list

This product includes:

1) User’s Manual (this Manual)····· 1 set
2) USB dongle for nanotagViewer ····· 1 set
3) CD-ROM for nanotagViewer ····· 1 disk

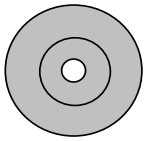
If there is anything missing, please contact us.



1) Users Manual



2) USB dongle



3) CD-ROM

1.3 System Requirements

You need a Windows PC and devices as follows.

PC	Windows PC(Follow the system requirements for your OS) which has 2 or more USB ports for USB dongle and NFC Card Reader No built-in NFC or Felica port
OS	Windows 11 Pro Windows 10 Pro (Not recommended due to end of Microsoft support)
Display	resolution of 1024 x 768 or greater
Other	SANKA nanotag® VSMN210
	FeliCa reader: Sony RC-S300 or RC-S380 (discontinued) Internet connection required for driver installation. *RC-S380 can be used as long as Sony driver support is available.
	Printer for printing Graphs

Caution



The PaSoRi NFC Card Reader (RC-S380) does not work on the PC with built-in NFC or Felica port.

1.4 nano tag specifications

Product Name	nano tag® (activity measuring device)
Model Name	VSMN 210
Detection method	3D acceleration and temperature sensors
Communication system	Near Field Communication (NFC system)
Communication distance	About 10 mm
Battery life	About 2 months *1
Operating	0 ～ 45 degrees Celsius
Measurement	10 ～ 45 degrees Celsius *2
Power supply ON/OFF	contactless communication system
Waterproof	Equivalent to IPX7
Dimensions	15 x 14.2 x 7.1 mm
Mass of the main body	about 2.7 g
Material of the main body	Case: ABS plastic
Built-in battery	CR1220 (Disposable, cannot be replaced)

*Recorded data will be erased if the nano tag is repaired or replaced.

1.5 Notes on using "nano tag"

➢ 1.5.1 Warranty Period of nano tag

1. The warranty period of nano tag is one year from the date of purchase.

➢ 1.5.2 Battery of nano tag

1. nano tag is disposable items.
2. Battery of nano tag cannot be replaced. The battery life is as follows;

Measurement	Battery Life	
Activity & Tempareture	60 days	*Recording whole day. *The connetion time with the PaSoRi NFC Card Reader is less than 2 minutes once a day.
Tempareture only	180 days	

Caution

Depending on the usage conditions, the battery life may be shorter than the above.
The measurement will stop when the battery is empty.

Caution

The battery will drain while the "nano tag" is waiting for the recording start by reservation of recording. In this case, it may not be possible to record for prescribed time.

➢ 1.5.3 Disinfection of the nanotag

Please see "3.2.3 Notes".

Caution

If you soak the nano tag in the antiseptic solution for more than the specified time, the strength of the case may decrease.

➢ 1.5.4 Handling Precautions

Prohibition

Please do not press strongly on the flat side of "nano tag".
It may cause damage.



Please do not press strongly on the flat side of "nanotag".

Prohibition



Never decompose.
Do not apply excessive vibration or shock.
It may cause malfunction.

➢ 1.5.5 Maintenance and Storage Instructions

Caution



Always keep clean. Wipe off the dirt on the body with a soft dry cloth.
Do not use benzene, thinner, etc. to remove dirt.
Do not store in the following locations.

- **A place where water splashes.**
- **Areas subject to high temperature, high humidity, direct sunlight, dust, or air containing salt.**
- **Areas subject to tilting, vibration or impact.**
- **Where chemical products are stored or where corrosive gases are generated.**

➢ 1.5.6 Safety Precautions

We are not responsible for the quality of the product if you do not follow the instructions or other proper usage instructions.

Warning



Keep nano tags out of reach of infants.
It may cause accidental ingestion or injury.
In case this product is swallowed, please consult a doctor immediately.

Prohibition



Do not put the nano tag in the fire. It may explode.

2 Install

This chapter describes the outline of software and hardware installation.
Follow the steps below to install and set up.

2.1	Preparation ····· p5
2.2	Installation of this software ····· p6
2.3	Insert a USB dongle into your PC····· p7
2.4	Installation and setup of NFC Port·····p8

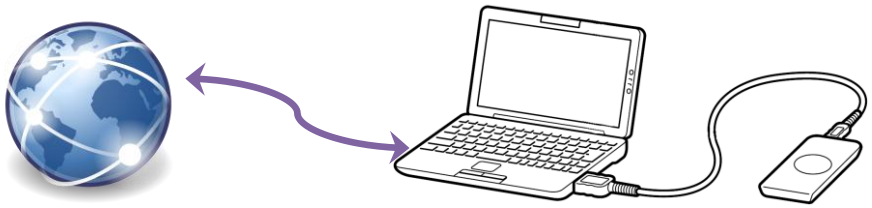
2.1 Preparation

This section describes how to install of software and hardware.
Please login to the PC with administrator privileges in Windows.
If you do not know, please check from [Control Panel]-[User Account].

Caution **Login with administrator in Windows**

It cannot install with "Standard User" or "Restricted Account".

In order to use "nano tag", you need to download and install the driver software for USB NFC port from the Sony website.



SONY FeliCa
https://www.sony.net/Products/felica/business/products/RC-S380.html

Caution

In order to use "nano tag", it is necessary to install the driver software for NFC port from SONY website.

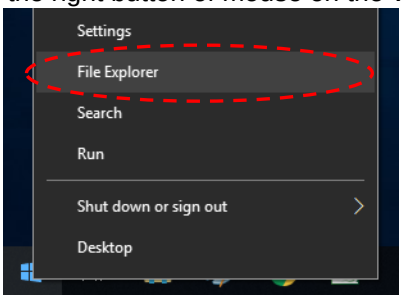
Caution

Please remove the USB dongle from your PC when you install the driver software for NFC port.

2.2 Installation of this software

This section describes how to install of "nanotagViewer".

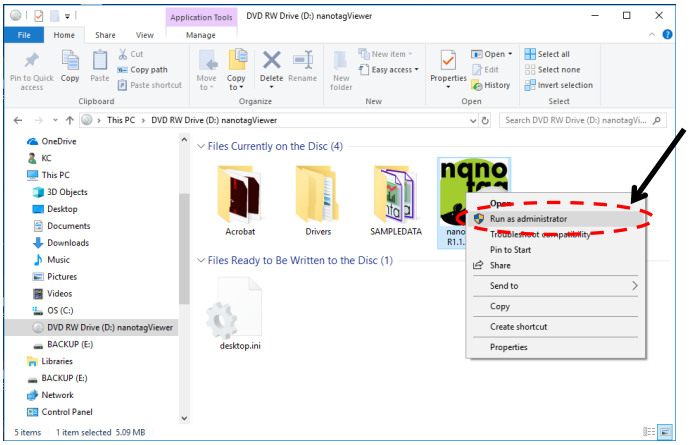
1. Insert CD of "nanotagViewer" into the CD drive.
2. Launch "File Explorer".
   (Click the right button of mouse on the Windows Start menu, or click the [File Explorer] icon)



or



3. Right-click "nanotagViewer Rx.x.x.xxxx.exe" in the CD, and select "Run as administrator".



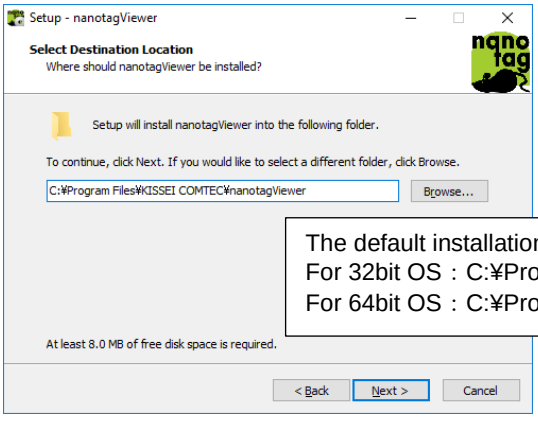
Select "Run as administrator" menu.

Caution

Right-click the icon for the installation program and select "Run As Administrator".

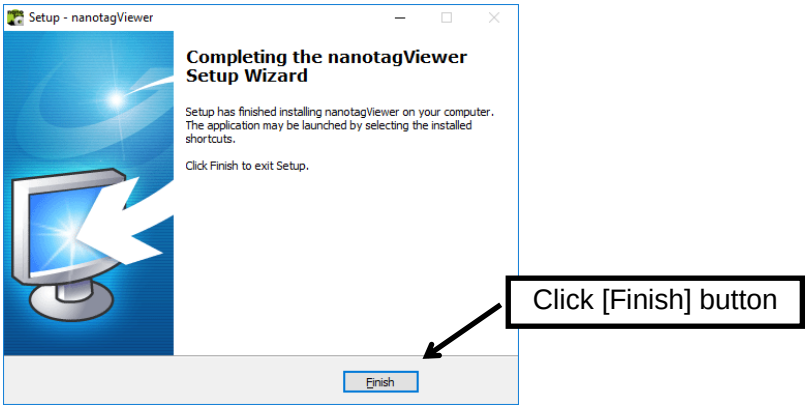
4. Installation will begin. Follow the messages displayed.

5. About installation destination, we recommend the "c:¥Program Files" folder for default installation destination.



The default installation folder is as follows.
For 32bit OS : C:¥ProgramFiles¥KISSEI COMTEC¥nanotagViewer
For 64bit OS : C:¥ProgramFiles (x86)¥KISSEI COMTEC¥ nanotagViewer

6. When the installation is complete, you will see the following screen.



2.3 Insert a USB dongle into your PC

This section describes how to insert of a USB dongle.
You need to insert a USB dongle into your PC In order to use "nanotagViewer".

for laptop：



for desktop：



Caution

Please insert the USB dongle to the PC after installing the program.

Windows will automatically recognize the USB dongle when you insert it into your PC.
In some cases, it takes 2-3 minutes.

2.4 Installation and setup of NFC Port

This section describes how to install and setup of NFC Port.
In order to read data from “nano tag” to your PC, it is necessary to set up NFC port.
First, install the driver software of NFC port and then its setup.

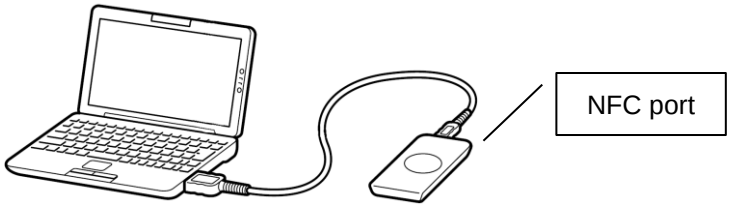
Order	[You need connect PC to the internet.]
2.4.1	Connecting to the Internet to download the Driver…p8
2.4.2	Connect PC and NFC port …… p8
2.4.3	Driver installation…… p8
2.4.4	NFC port settings ……p10
3.3.1	Check status…… p21 Check that the NFC port connect correctly.

2.4.1 Connect to the internet to download the driver

Download the FeliCa reader driver on a computer connected to the internet.
Internet connection is required for the initial setup only.
Please download the latest "Basic Software (NFC Port Software)" from the Sony website.
(We cannot provide it due to redistribution restrictions)
https://www.sony.co.jp/Products/felica/consumer/support/download/
[As of September 2025]
Select your device model on this page.
Once the driver is installed and data can be read, internet connection will no longer be required.

2.4.2 Connect PC and NFC port

Connect an NFC port to the USB port of your PC.



2.4.3 Driver installation

The driver will be installed automatically when you insert NFC port to your PC in windows7 OS.

However, Windows 10 and some windows 7 do not automatically start driver installation.
In order to use NFC port, you have to manually install “NFC Port Software”.

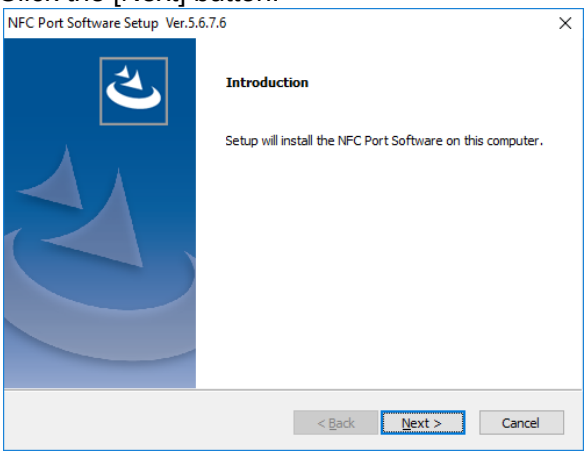
Caution

In order to use “NFC port”, you have to manually install “NFC Port Software”. Please see to the next page for the installation method.

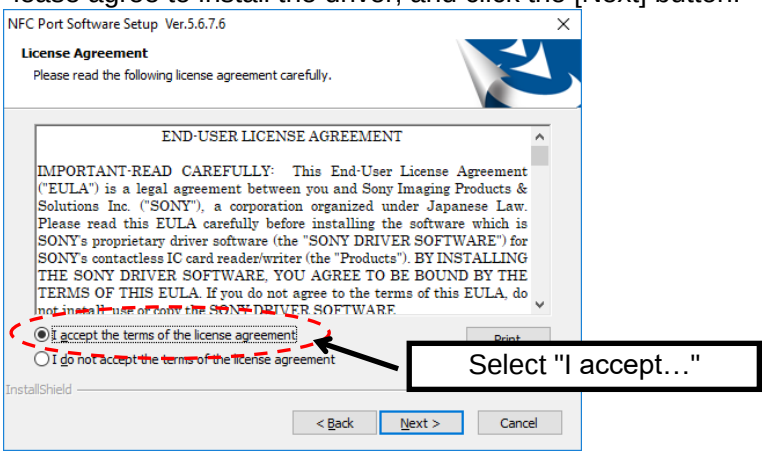
1. Please download the “NFC Port Software” from the Sony website and run it.

   SONY FeliCa
   https://www.sony.net/Products/felica/business/products/RC-S380.html

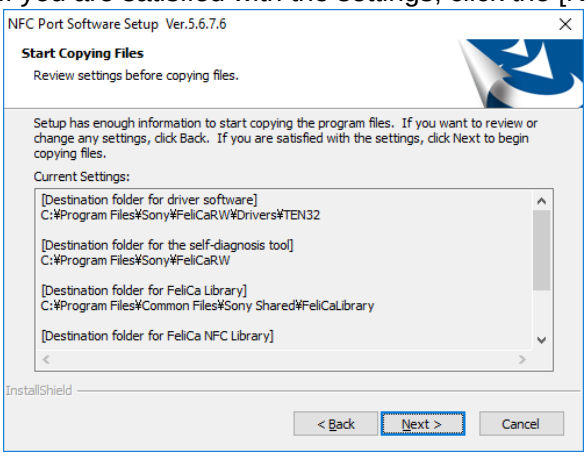
2. Click the [Next] button.



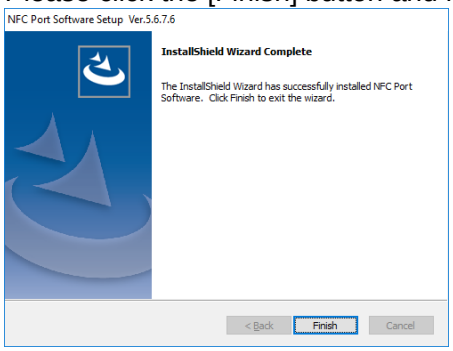
3. Please agree to install the driver, and click the [Next] button.



4. If you are satisfied with the settings, click the [Next] button to begin installing files.



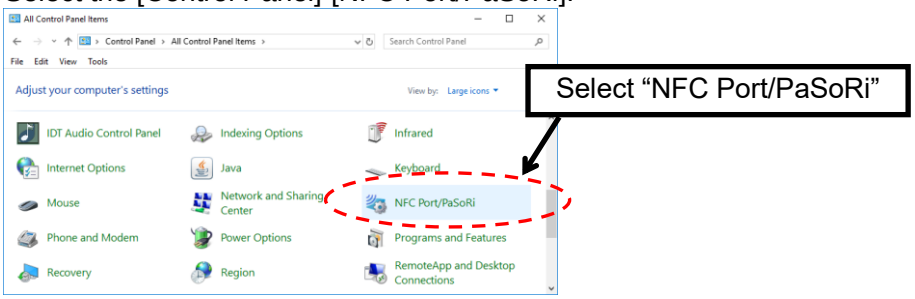
5. The following screen will be displayed when installation is complete.
   Please click the [Finish] button and restart your PC.



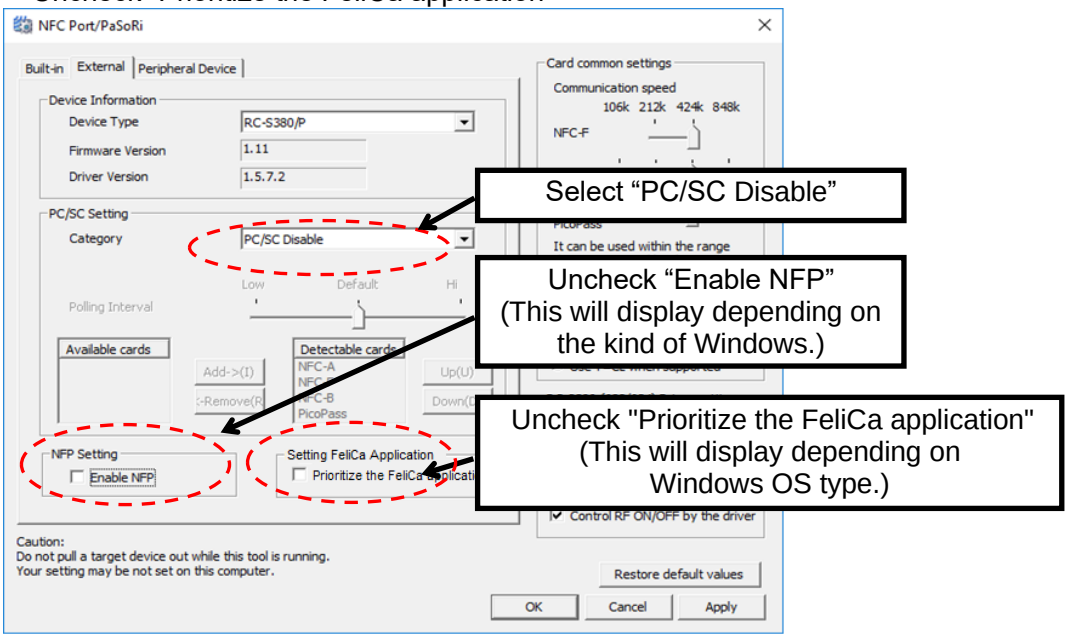
2.4.4 NFC port settings

You will need setting of "NFC Port/PaSoRi" in the "control panel", when using NFC port.

1. Select the [Control Panel]-[NFC Port/PaSoRi].



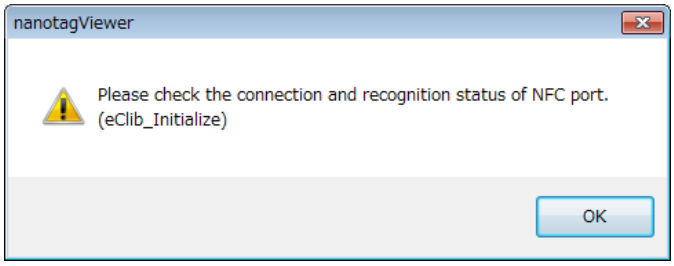
2. Select "External" tab and set as follows.
   ・Select “PC / SC Invalid"　・Uncheck “Enable NFP”
   ・Uncheck “Prioritize the FeliCa application”



3. Click the [Apply] button and then the [OK] button.
   Be sure to restart your PC.

2.4.5 When NFC port is not recognized

"nanotagViewer" may display the following message when you start it and select "nano tag operation list" menu.



In this case, Windows does not recognize NFC port (RC-S380/S). You need to reconnect to the Internet and re-install the driver of it.

If you cannot connect to the Internet on your PC, download the driver software from another PC connected to the Internet.

SONY FeliCa
https://www.sony.net/Products/felica/business/products/RC-S380.html

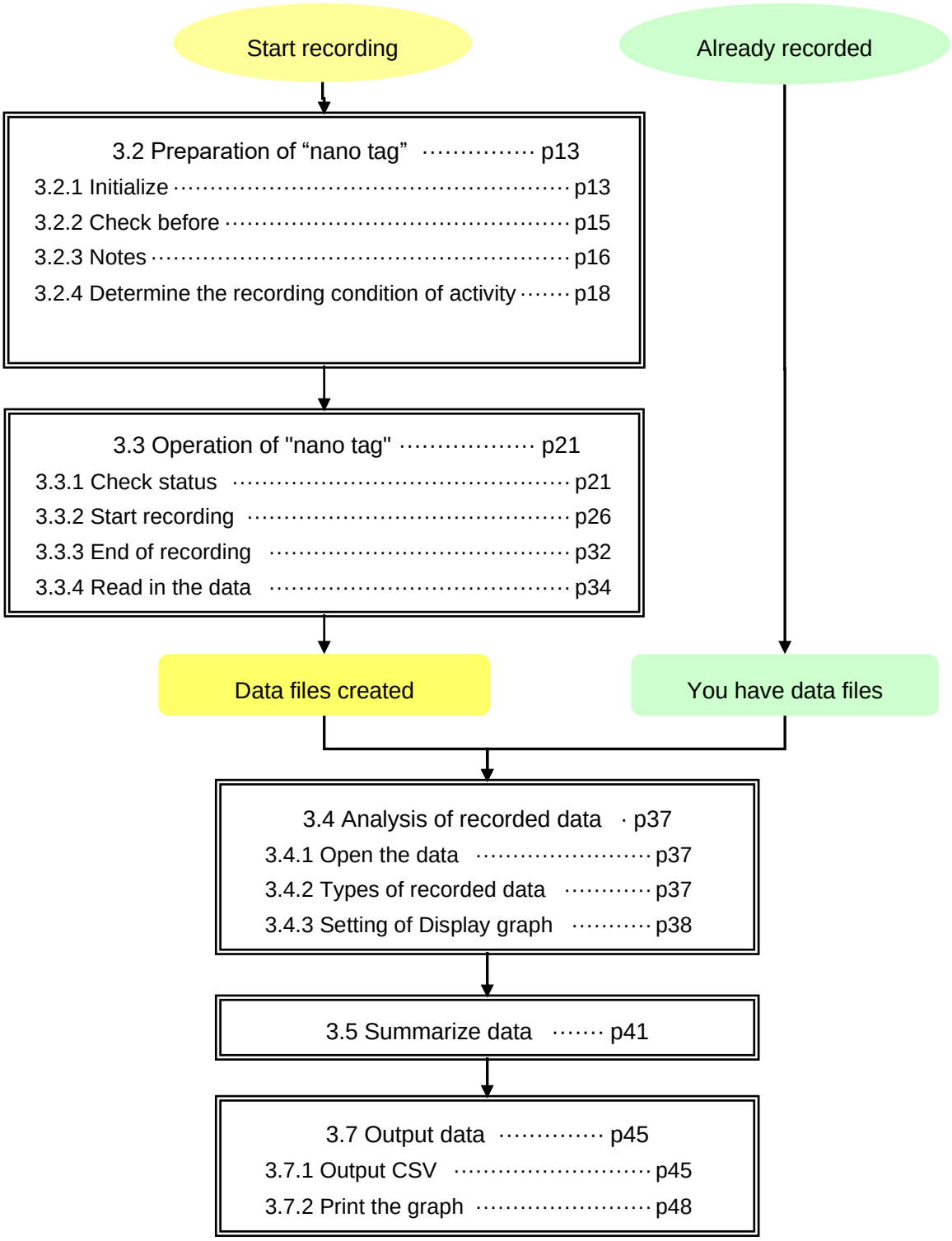
Please download the "NFC Port Software".

3 Operation

This chapter outlines the general process of using “nanotagViewer” software.
Please see it as a clue to decide the operation method.

3.1 Flow of operation

The flow of operation is as follows.



3.2 Preparation of “nano tag”

This section describes how to start recording with "nano tag".
Please see to the instruction manual attached to the device for precautions on the "nano tag" themselves.

3.2.1 Initialize

If you need to initialize as follows, please run in the order of "Check status"-> "initialize".
・You use “nano tag” for the first time.
・Change the subject and record data.

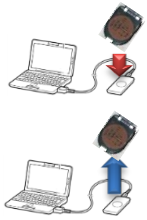
Caution

All data will be lost when you initialize "nano tag".
If you are using "nano tag", please read the data first and then initialize it.

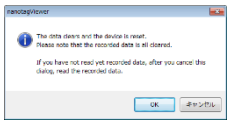
1) Select the [Data]-[List of operating nano tag] menu, and then click the [Confirm condition] button.



2) Please place “nanotag” on the NFC port, and if the screen appears, release it.

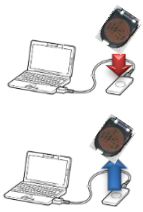


3) After click the [Initialization] button on the status check screen, place "nanotag" on the NFC port again.



All data in the “nanotag” will be erased.

4) Please place “nanotag” on the NFC port, and if the completion screen appears, release it.



Caution

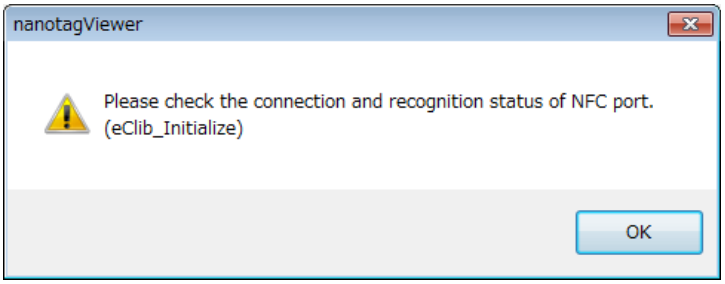
In order to run NFC communication with the "nano tag", you are necessary to move "nano tag" close to the NFC port (within 1cm) in the place where there is no metal around.

<More>

1. Connect NFC port and your PC with USB cable.
   About how to connect, please see to the “2.4.2 Connect PC and NFC port (p8)” section.

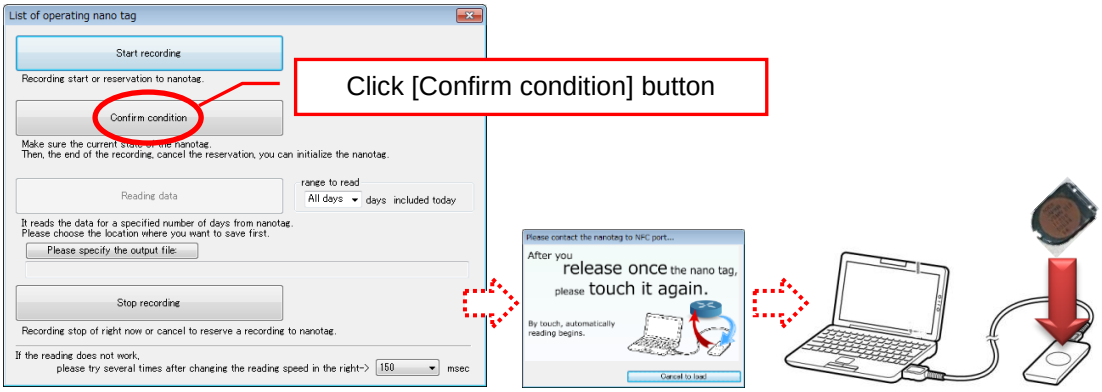
2. Start "nanotagViewer"
   First, run the software by double-clicking the "nanotagViewer" icon.
   Next, select the [Data]-[List of operating nano tag] menu.

   If the following screen appears, NFC port driver is not installed correctly.
   Please see to the “2.4.5 When NFC port is not recognized (p11)” section.
   Or, you may not have set up the NFC port, so please see to “2.4.4 NFC port settings (p10)” section.

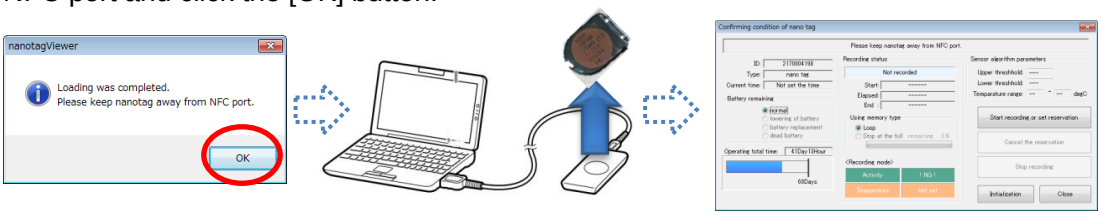


3. Click the [Confirm condition] button, and move "nano tag" closer to the NFC port.

   Move the antenna side (the surface without the text written) of the "nano tag" close to the NFC port.



   When the screen of "Loading was completed" appears, please release "nano tag" from NFC port and click the [OK] button.



   First, make sure that "Recording status" column at the center of the "Confirming condition of nano tag" screen is "Not recorded".
   -> If "Recording" or "Reserved" display in "Recording status" column, you need to end the recording or cancel the reservation.
   For more information on the "Confirming condition of nano tag" screen, please refer to the following section.

4. Click the [Initialization] button, and move "nano tag" closer to NFC port.

   Move the antenna side (the surface without the text written) of the "nano tag" close to NFC port.



When the screen of "It made a device reset" appears, it is complete.

Please release "nano tag" from NFC port.
For 10 to 20 seconds after the initialization, "nano tag" does not accept communication.

Caution

Please run “Initialization” only when start recording with a different subject, as all data will be lost.

3.2.2 Check before

Check before when "nano tag" is implanted in animals.

1. Please initialize the "nano tag" when recording in a new subject.

   Even if recorded data in the past still remains in the "nano tag", the data will not be available when recording is newly start.

2. Be careful of the battery level.

   If "Battery remaining" in the "Confirming condition of nano tag" screen is other than "normal", the battery will soon be depleted. Please note that you will not be able to use it soon.

   The "nano tag" will keep the battery for about 60 days (measurement time including activity) or 180 days (temperature only mesurement). However, frequent readings 2-3 times a day or leaving the "nano tag" on NFC port can cause the battery to run out quickly and run out within prescribed time.

Caution About battery life

The battery life may be less than prescribed time depending on the communication situation.

3. When you implanted the “nano tag”and record the data for a long time.

・ Please read or retrieve the data before the memory of "nano tag" is full.
The number of days that can be stored in "nano tag" differs depending on the recording conditions (data resolution). The “nano tag” cannot save data more than the number of upper limit days.

Therefore, make sure that you read the data within the upper limit days.
When you read the data, you will get one file.

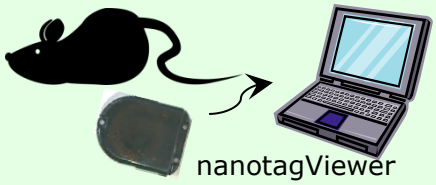
Repeat this process until you reach the desired recording time.
How to combine multiple files is as follows.
・Use the "Combine recording files" function of "nanotagViewer".
・Export to CSV files and combine on Excel

If you set the recording condition "Using memory type" to “Loop".



The data will be overwritten from the beginning when the memory is full.

Therefore, before memory is full.



Retrieve the data and save it to a file before the memory is full. And then after, repeat this process.

3.2.3 Notes

Notes when "nano tag" is implanted in animals.

1. Disinfect "nano tag" by the specified method.
For instructions on disinfecting, please refer to the instruction manual supplied with the “nano tag”.

・Immersion time in povidone iodine and ethanol: within 10 minutes
* Use povidone iodine solution (isodine solution 10%) and dilute to a concentration of 2% or less.

・Immersion time to Chlorhexidine gluconate and Osval fluid: within 30 minutes
* Use Chlorhexidine gluconate (5% hibitene solution) and dilute to a concentration of 0.5% or less.
* Use 10% of Osban solution and dilute to a concentration of 2% or less.

2. The antenna side of “nano tag” outward

Implant in the body so that the antenna side of the "nano tag" (surface without written characters) is on the outside. (See the figure below)

Antenna side



No characters

Back side



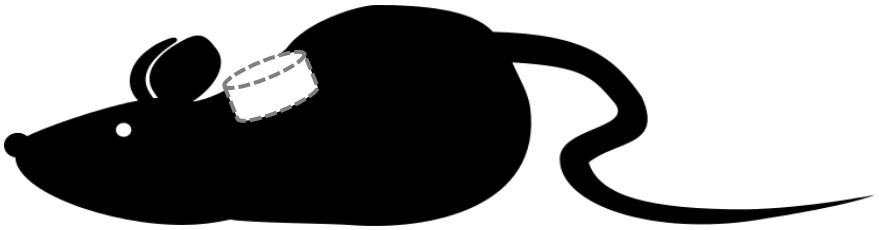
You can see characters such as ID.

3. Implant subcutaneously

The range in which NFC port can communicate is 1 cm without obstacles. Therefore, please be sure to implant subcutaneously.
If you implant "nano tag" in the body such as the abdominal cavity, you cannot read the data unless it is removed from the body.

For reference, it seems that it takes about 3 days for recovery when "nano tag" is implanted in the neck or abdomen of a rat or mouse.

Turn the antenna side of the "nanotag" outward.



4. Communicate with "nano tag" before adhesion.

When the "nano tag" implanted subcutaneously in rats, adhesions may begin for about a week.
Please run "Confirm condition" of "nanotagViewer" appropriately to check whether the antenna side of the "nano tag" is facing outside the body.

If there is a space around the "nano tag" implanted under the skin, there seems to be a case where the "nano tag" rotates. In this case, the communication situation becomes worse. Please slowly invert the "nano tag" so that the antenna side faces outward.

3.2.4 Determine the recording condition of activity

You can change the threshold for calculating the amount of activity (number of vibrations per time resolution).

> Our company recommended values are available for subcutaneous injection into the neck or abdomen of mice and rats.
> Please see “4.1 Calculation method of Activity (cross counts) (p49)” section for activity calculation.

If you implanted "nano tag" other than subcutaneous in the neck or abdomen, you need to change the threshold. After implanting "nano tag" in your desired position, record the composition of acceleration in 3 axes of the case with strong motion and the case with quiet, and consider what threshold value should be set.

Info



When measuring under the following conditions,
You need to consider the recording conditions (Threshold for calculating the activity).

- Other than the neck or abdomen of a rat or mouse
- Buried in animals other than rats and mice
- When you want to measure movements other than circadian rhythm

This section illustrates the method used to determine our company recommended value.
To measure circadian rhythm by placing a nanotag under the skin of the abdomen, we examine the threshold.

1) Record vector data (composition of acceleration in 3 axes)

The vector wave can be recorded for up to only 27 minutes, so it is important to plan the assumption in advance and the readout timing.
Video recording equipment was used for simultaneous monitoring.

1. reservation on nanotag to start recording in a week

   The recording conditions were "vector wave", "loop", "Start recording after 1 week", and "56 days recording".

2. nanotag was implanted subcutaneously in the abdomen and recovered for a week.

3. Vector waves in three motions are recorded in order.

   A: Quiet and not moving (stay in sleep)

   We made sure that the animal was sleeping on a video monitor during the light period when it was sleeping frequently.

   B: It's moving a little (waking up and mumbling)

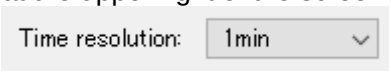
   We continued to check the video monitor to see when the transition from sleep to wakefulness is coming. About 5 minutes after waking up, we read the data.

   C: Awakening and moving around (a state of movement as in exploratory activity)

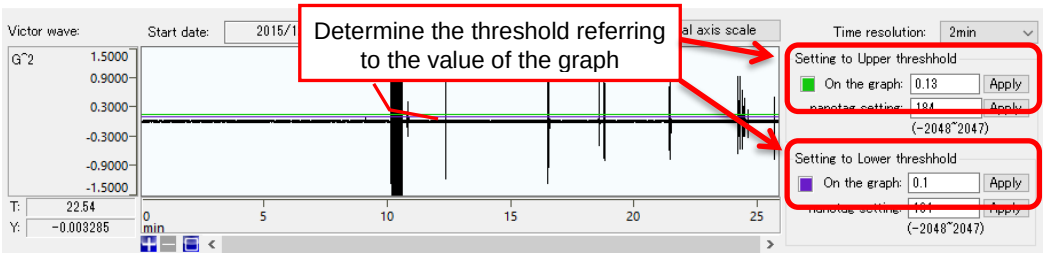
   We kept recorded until dark period and read the data while watching the animal on the video monitor, immediately after it was exploring.

2) Open the recorded data and review the thresholds

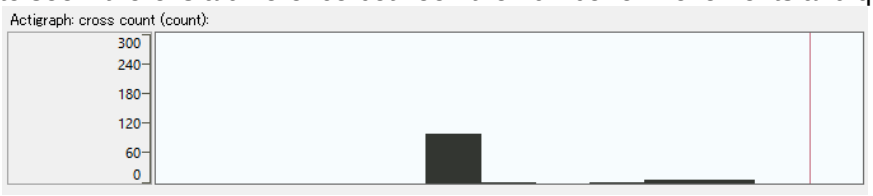
1. Open the data of composition of acceleration in 3 axes, and select the [Time resolution] at the upper right of the screen.



2. Input the upper and lower threshold values with reference to the vertical axis value of the Vector wave [composition of acceleration in 3 axes] at the top of the screen, and click the [Apply] button. The values can reference by moving the mouse.



3. You should check the "Actigraph (Cross count)" displayed at the bottom of the screen to see if there is a difference between the number of movements and quiet of animals.

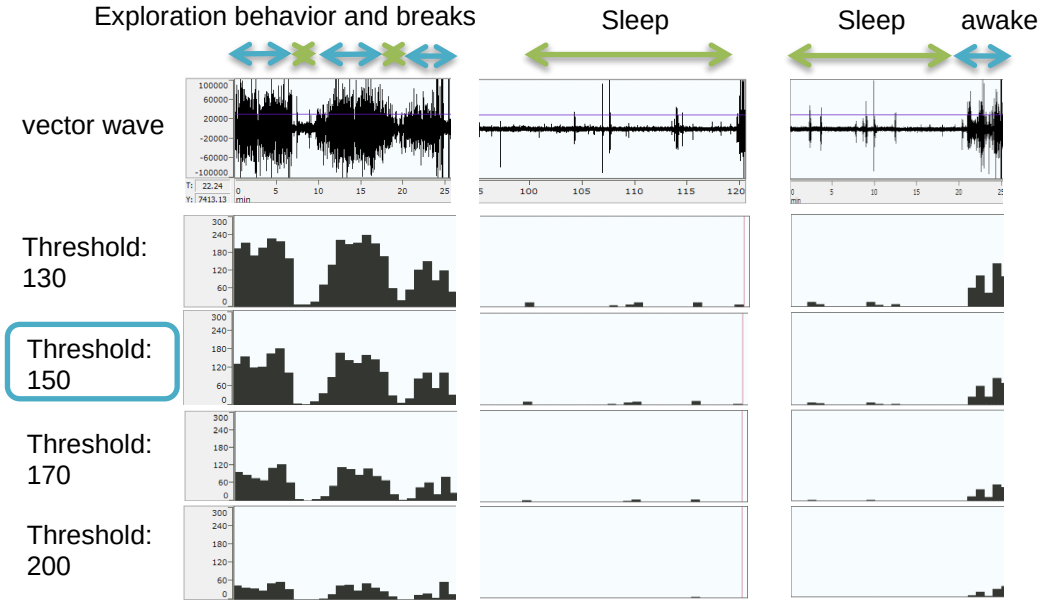


4. The value of "nano tag setting" is displayed in the upper right of the screen is the threshold value.



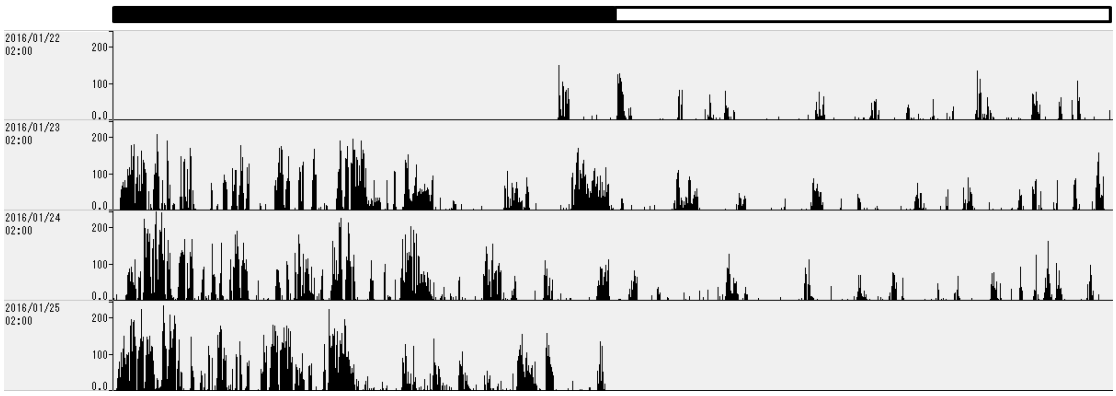
These are the rooted values on the graph.

The actual data looked like this:



We first set the upper threshold equal to the lower threshold and apply 150.

3) Measure activity using the thresholds considered



We decided to use a threshold value of 150 because we could confirm that there was a difference in the amount of activity between the light and dark periods and that we could grasp the circadian rhythm.

3.3 Operation of "nano tag"

This section describes how to start and end recording of "nano tag", read data and check the status.

3.3.1 Check status

You can check the current condition of "nano tag".
This function can also be used to check whether the recording start and end operations described below have been completed successfully.

1. Connect NFC port to your PC

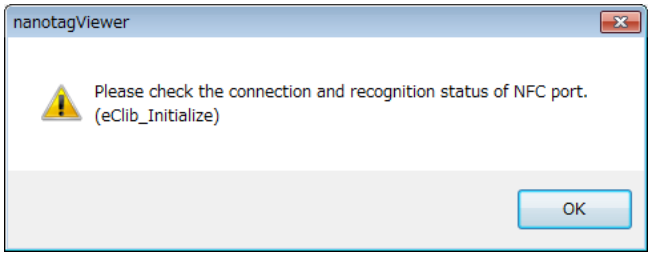
   For information on connecting NFC port to your PC, please see "2.4 Installation and setup of NFC Port (p8)" section.

2. Start "nanotagViewer" and select the [Data] - [List of operating nano tag] menu.

   At this point, if the screen below appears, NFC port is not connected to your PC.
   Restart your computer, and then reconnect NFC port.

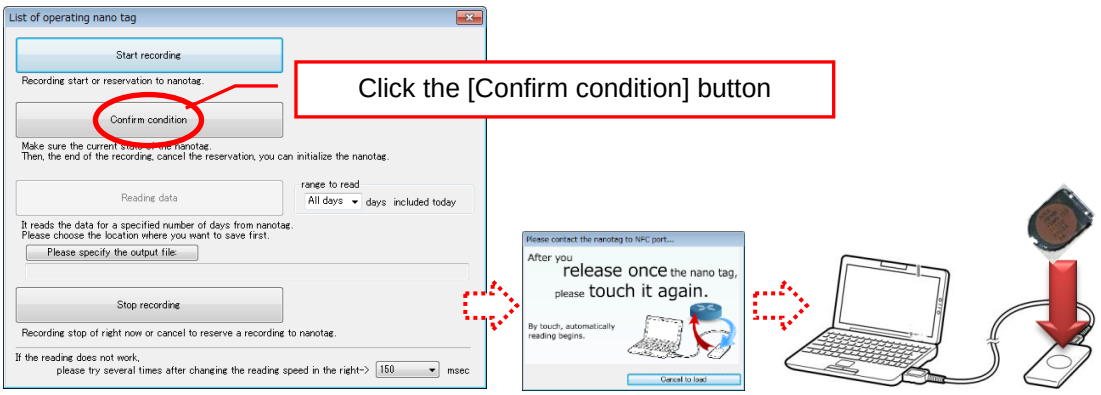
   Or, NFC port driver may not be installed. Please see to "2.4.5 When NFC port is not recognized (p11)" section and reinstall.

   Also check your NFC port settings "2.4.4 NFC port settings (p10)" section.

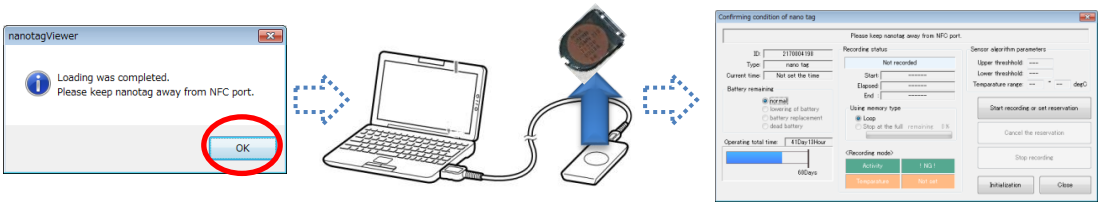


3. Click the [Confirm condition] button, and to move "nano tag" closer to NFC port.

   Move the antenna side (the surface without the text written) of "nano tag" close to NFC port.



When the screen of "Loading was completed" appears, release "nano tag" from NFC port and click the [OK] button in the screen.

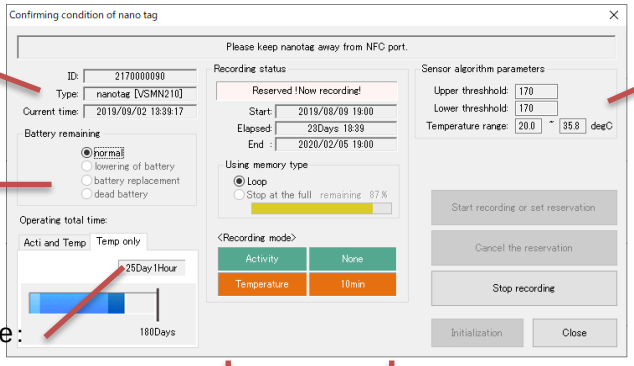


4. About the [Confirm condition] screen

Equipment ID:
The serial number of "nanotag".

The estimated battery life is 60 days.

Operating total time:
Displays the total number of recording days and reserved waiting days.



Algorithm:
Threshold value set at the start of recording.
The temperature displays the measurable range.

Recording mode:
See to the explanation as follows.

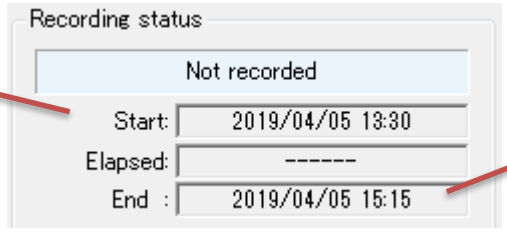
Recording mode:
・Recording status ・・・ Current condition of "nano tag"

Status	Display	Period displayed in "Recording status"	Are each status included in "Operating total time"?
Not recorded	Not recorded	Period of the latest recorded data saved in "nano tag"	× Not included
Reserved	Reserved	Reserved period	✓ Included (*1)
Now recording	!Now recording!	Current recording period	✓ Included
Now recording by reservation	Reserved !Now recording!	Current recording period and reservation end time	✓ Included

*1: Including about 35% of the time during recording.

1) Not recorded
This status indicates that you can read previously recorded data.
You can start recording or make a reservation for recording now, too.

Start:
Start date and time of the data saved in the "nanotag".

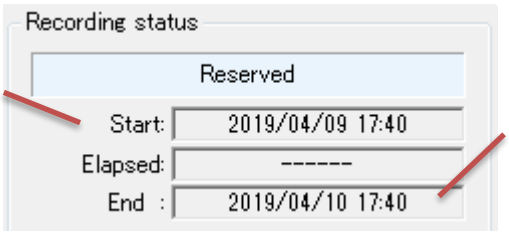


End:
End date and time of the data saved in the "nanotag".

★In this case, it is not included in "Operating total time".

2) Reserved
This status indicates that the "nano tag" is in reservation recording, and it waits until reservation time. (If you want to record it now, please cancel the reservation and start recording.)

Start reservation:
Recording start date and time reserved for "nanotag".



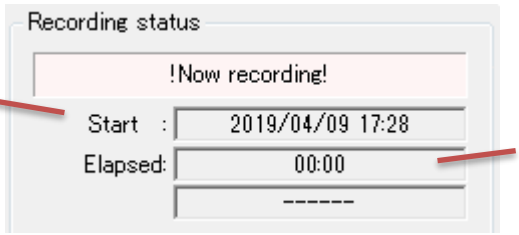
End of reservation:
Recording end date and time reserved for "nanotag".

★The number of days until the reserved recording starts is also added to "Operating total time". The battery is exhausted even while waiting for recording. The power consumption is about 35% compared to the time of recording. It is added to "Operating total time" according to the percentage.

For example, if you reserve to start recording after 1 day, 8 hours will be added to "Operating total time". (24 hours x 35%)
Therefore, if you use timer reservation, the number of days in "Operating total time" is larger than the number of days actually recorded. And, if the timer reservation wait time is long, you may not be able to record for 60 days.

3) Now recording
This status indicates that the "nano tag" is recording.
If "Using memory type" is selected for "Loop".
・Memory is not full:

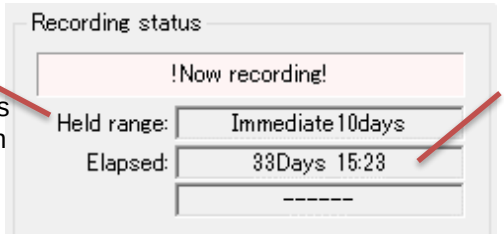
Start:
Date and time recording started.



Elapsed time:
Time elapsed since recording started.

・Memory is full and the first data is overwritten:

Held range:
The number of days of data remaining in "nanotag". (Include today).



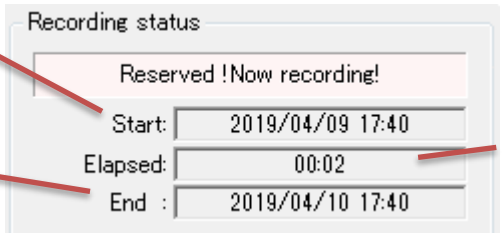
Elapsed time:
Indicates the time elapsed since the memory was full. (As the memory is full, the actual recording start time is

★Of course, the recording period is added in "Operating total time".

4) Now recording by reservation
This status indicates that the "nano tag" is recording by reservation recording.
If "Using memory type" is selected for "Loop".
・Memory is not full:

Start:
Date and time recording started.

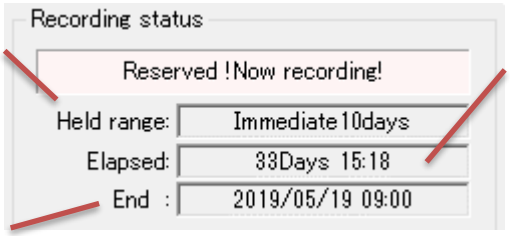
End:
Date and time to end recording.



Elapsed:
Time elapsed since recording started.

・Memory is full and the first data is overwritten：

Held range：
Indicates how many days (including today) of data remain in memory because the memory is full.



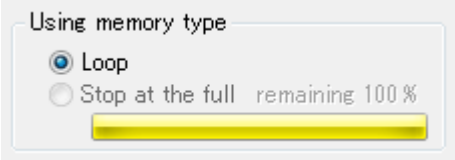
Elapsed time：
Indicates the time elapsed since the memory was full. (As the memory is full, the actual recording start time is lost.)

End：
Date and time to end recording.

★Of course, the recording period is added in "Operating total time".

・Using memory type

Select control method when the memory is full.



[Loop]…

When the memory is full, the oldest data is overwritten.
For example, if the memory is full after 3 days, the data obtained by recording will be the "latest 3 days".

[Stop at the full]…

When the memory is full, recording stops automatically.
For example, if the memory is full after 3 days, the data obtained by recording will be the "3 days from the start of recording".

・Recording mode

It indicates the time resolution of the data.

The number of days that can record is as follows.

		Activity						
	Time resolution	12sec	30sec	1min	2min	4min	5min	No record
Temperature	12sec	4	-	-	-	-	-	9
	30sec	6	10	-	-	-	-	22
	1min	7	13	20	-	-	-	45
	2min	8	16	25	37	-	-	60/ 90 #
	4min	8	18	30	47	60	-	60/180 #
	5min	8	18	31	50	60	60	60/180 #
	10min	-	-	-	-	-	-	--/180 #
	No record	20	20	36	60	60	60	-

UOM: Days

★ The 10-minute resolution of temperature can be recorded with the model VSMN 210 or later.

Only model VSMN 210 and later can record to 90 or 180 days.
VSMN 200 can record to 60 days.

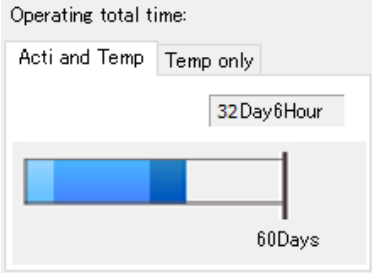
★"No record" means that neither activity nor temperature is recorded.

★The limitation of the number of days in the above table is due to the memory capacity and the battery life of the "nano tag".

Operating total time:

The accumulated time during recording and reservation standby is displayed.

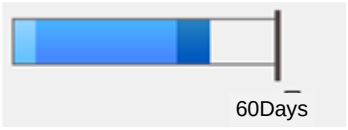
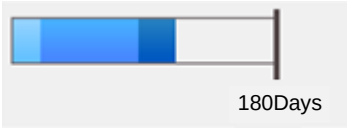
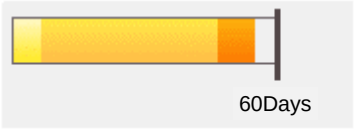
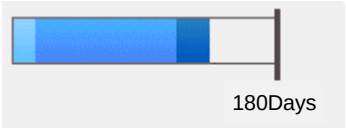
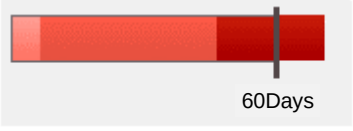
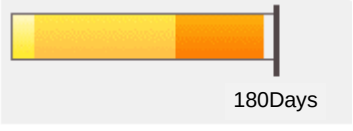
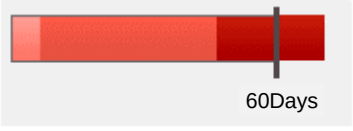
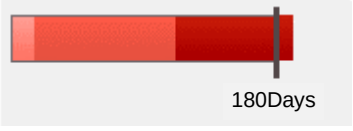
For model VSMN 210 and later, the power consumption differs between "a measurement involving momentum" and "Measure temperature only".
If you want to perform "Temperature only" from now on, refer to the [Acti and Temp] tab. If you want to measure "a measurement involving momentum", refer to the [Temp] tab.



The battery is exhausted even while waiting for recording. The power consumption is about 35% compared to the time of recording. It is added to "Operating total time" according to this percentage.
For example, if you reserve to start recording after 1 day, 8 hours will be added to "Operating total time". (24 hours x 35%)
Therefore, if you use timer reservation, the number of days in "Operating total time" is larger than the number of days actually recorded.

・Bar Charts

Dark color ・・・Accumulated recording time measured only temparature
Medium dark color・・・Accumulated recording time measured including activity
Light color ・・・Accumulated reserve wait time
(=Calculated as "Consume about 35% of the recording")

Bar Charts		State of battery consumption
[Acti and Temp] tab	[Temp] tab	
		There is enough battery power. Please see as a guide for how many days worth of data the "nano tag" can record.
		Battery is low. The cumulative number of days exceeds 50 or 150. If you frequently communicate with the NFC port, the battery may run out within 60 or 180 days.
		
		Battery is very low. The cumulative number of days exceeds 60 or 180. Do not use "nano tag" in this state.

3.3.2 Start recording

You can set the recording conditions and start recording now or make a recording reservation.

If you start recording or reserve recording, all data in "nano tag" will be deleted.

Caution

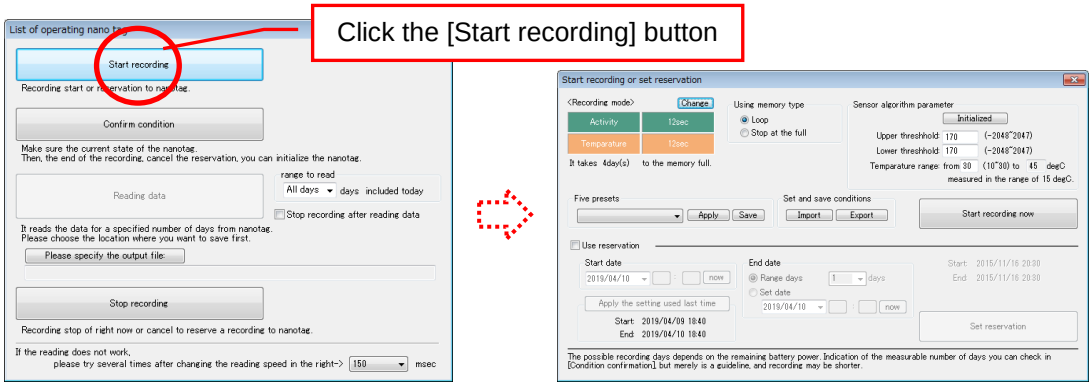
If you start recording or reserve recording, all data in "nano tag" will be deleted. Therefore, if data previously recorded in "nano tag" remains, be sure to read it first.

➢ Start recording right now

Start recording "nano tag".

1. After connecting NFC port to your PC, run "nanotagViewer" and select [Data] - [List of operating nano tag] menu.

2. Click the [Start recording] button to set the recording conditions.



"nanotagViewer" comes with 4 typical recording conditions (No.1~4) for rats and mice.
1) Select one from 4 recording conditions.
2) Click the [Apply] button to set the recording conditions.

・About ”Using memory type”

[Loop]…

When the memory is full, the oldest data is overwritten.
For example, if the memory is full after 3 days, the data obtained by recording will be the "latest 3 days".

[Stop at the full]…

When the memory is full, recording stops automatically.
For example, if the memory is full after 3 days, the data obtained by recording will be the "3 days from the start of recording".

・About ”Recording mode”

If you want to change the "Recording mode", click the [Change] button and select the time resolution of the activity or temperature.

The number of days that can record is as follows.

		Activity						
	Time resolution	12sec	30sec	1min	2min	4min	5min	No record
Temperature	12sec	4	-	-	-	-	-	9
	30sec	6	10	-	-	-	-	22
	1min	7	13	20	-	-	-	45
	2min	8	16	25	37	-	-	60/ 90#
	4min	8	18	30	47	60	-	60/180#
	5min	8	18	31	50	60	60	60/180#
	10min	-	-	-	-	-	-	- /180#

	No record	20	20	36	60	60	60	-

UOM: Days

★ The 10-minute resolution of temperature can be recorded with the model VSMN 210 or later.

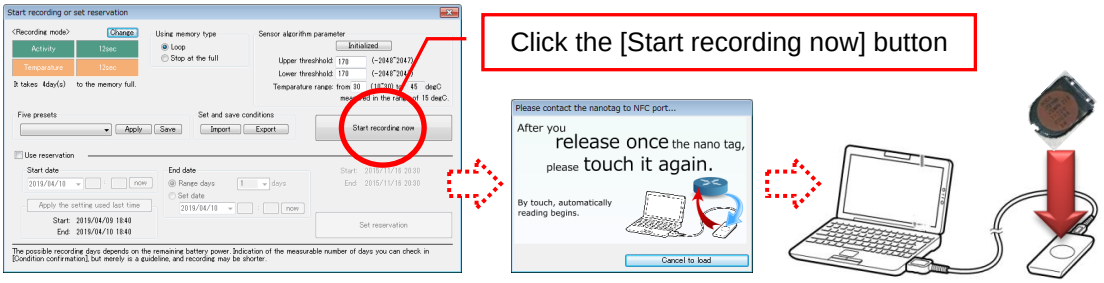
Only model VSMN 210 and later can record to 90 or 180 days.
VSMN 200 can record to 60 days.

★"No record" means that neither activity nor temperature is recorded.
★The limitation of the number of days in the above table is due to the memory capacity and the battery life of the "nano tag".
★"nanotagViewer" can record XYZ raw wave and Vector wave (composition of acceleration in 3 axes).

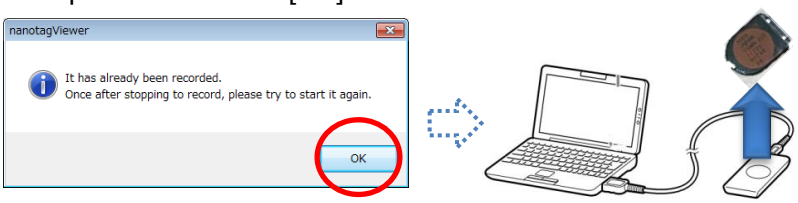
For example, if you set the time resolution of activity and temperature to 1 minute, you can record up to 20 days. In this case, data is recorded every minute.

If you read the data in a timely manner before the memory is full, you can record up to 60 or 180 days. If you want to record for a long time continuously like this, set "Using memory type" to "Loop".

3. Click [Start recording now] button to move "nano tag" close to the NFC port.



When the screen of "It has already been recorded" appears, release "nano tag" from the NFC port and click the [OK] button.



➢ Reserve for recording

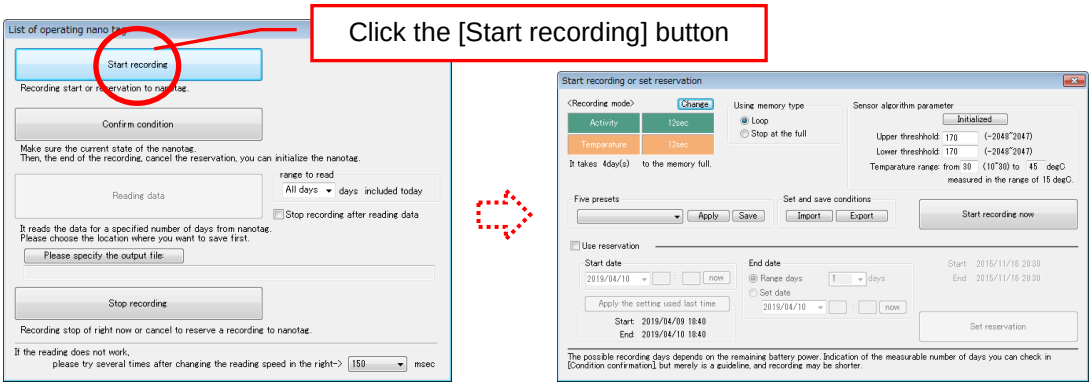
Set scheduled recording with "nano tag".

★The number of days until the reserved recording starts is also added to "Operating total time". The battery is exhausted even while waiting for recording. The power consumption is about 35% compared to the time of recording. It is added to "Operating total time" according to the percentage.

For example, if you reserve to start recording after 1 day, 8 hours will be added to "Operating total time". (24 hours x 35%)
Therefore, if you use timer reservation, the number of days in "Operating total time" is larger than the number of days actually recorded. And, if the timer reservation wait time is long, you may not be able to record for prescribed days.

1. After connecting the NFC port to your PC, run "nanotagViewer" and select the [Data] - [List of operating nano tag] menu.

2. Click the [Start recording] button to set the recording conditions.



"nanotagViewer" comes with 4 typical recording conditions (No.1~4) for rats and mice.
1) Select one from 4 recording conditions.
2) Click the [Apply] button to set the recording conditions.

・About ”Using memory type”

[Loop]…

When the memory is full, the oldest data is overwritten.
For example, if the memory is full after 3 days, the data obtained by recording will be the "latest 3 days".

[Stop at the full]…

When the memory is full, recording stops automatically.
For example, if the memory is full after 3 days, the data obtained by recording will be the "3 days from the start of recording".

・About ”Recording mode”

If you want to change the "Recording mode", click the [Change] button and select the time resolution of the activity or temperature.

The number of days that can record is as follows.

		Activity						
	Time resolution	12sec	30sec	1min	2min	4min	5min	No record
Temperature	12sec	4	-	-	-	-	-	9
	30sec	6	10	-	-	-	-	22
	1min	7	13	20	-	-	-	45
	2min	8	16	25	37	-	-	60/ 90#
	4min	8	18	30	47	60	-	60/180#
	5min	8	18	31	50	60	60	60/180#
	10min	-	-	-	-	-	-	- /180#
	No record	20	20	36	60	60	60	-

UOM: Days

★ The 10-minute resolution of temperature can be recorded with the model VSMN 210 or later.

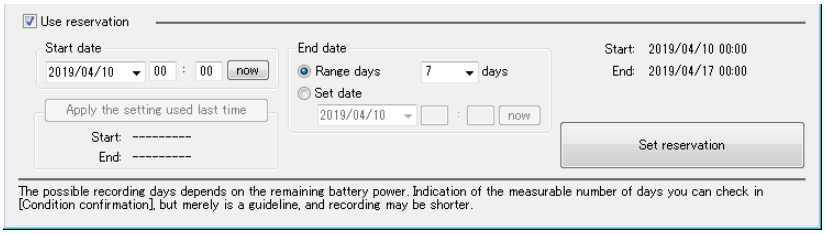
Only model VSMN 210 and later can record to 90 or 180 days.
VSMN 200 can record to 60 days.

★"No record" means that neither activity nor temperature is recorded.

★The limitation of the number of days in the above table is due to the memory capacity and the battery life of the "nano tag".

★"nanotagViewer" can record XYZ raw wave and Vector wave (composition of acceleration in 3 axes).

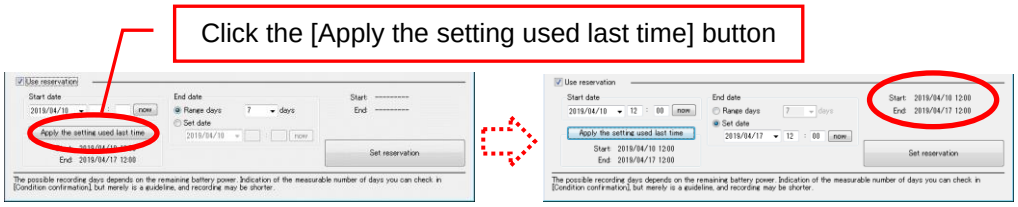
3. Check the [Use reservation] and set the start and end date and time.



To record until the memory of "nanotag" is full:
First set [Start Date] and then [End Date] set a number of days that is more than the memory is full, and by setting [Using memory type] to "Stop at the full", you can record fill the memory with data.

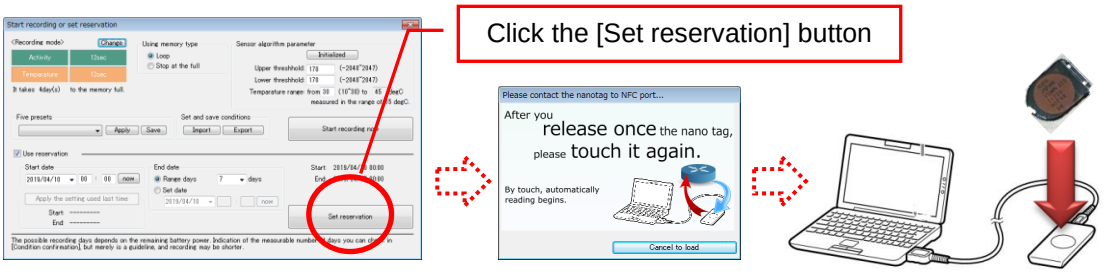
Info

In recording reservation, you can use the [Apply the setting used last time] function to instantly set the date and time of the previous reservation. This is useful if you want to record several "nano tag" at the same time.
(Be sure to check the start and end times.)

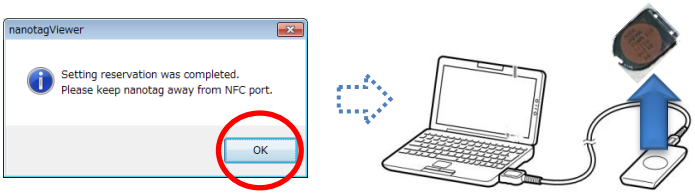


4. Click the [Set reservation] button to move "nano tag" close to NFC port.

   Move the antenna side (the surface without the text written) of "nano tag" close to the NFC port.



   When the screen of "Setting reservation was completed." appears, release "nano tag" from the NFC port and click the [OK] button.



3.3.3 End of recording

Stop recording or cancel the reservation of "nano tag".

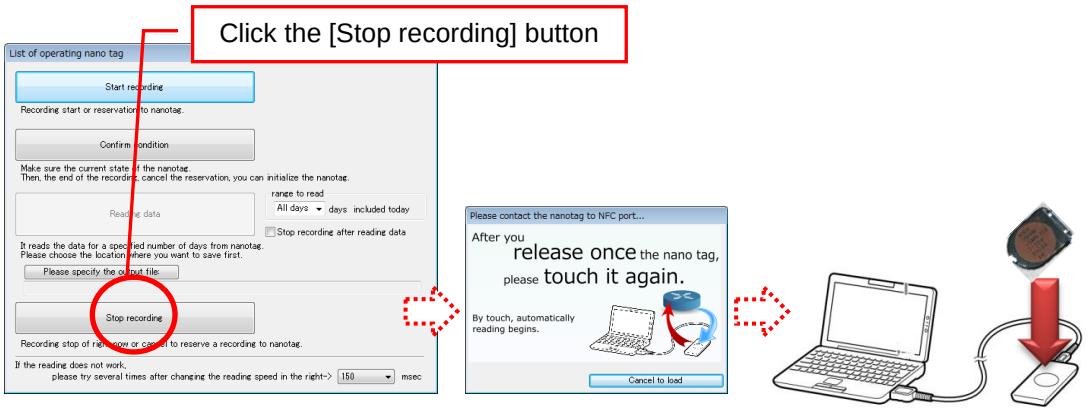
➢ End recording right now

Stops recording "nano tag".

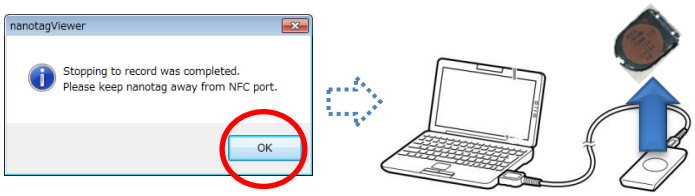
1. After connecting your PC to NFC port, run "nanotagViewer" and select the [Data] - [List of nano tag operations] menu.

2. Click the [Stop recording] button to move "nano tag" close to the NFC port.

   Move the antenna side (the surface without the text written) of "nano tag" close to the NFC port.



When the screen of "Stopping to record was completed." appears, release "nano tag" from the NFC port and click the [OK] button.



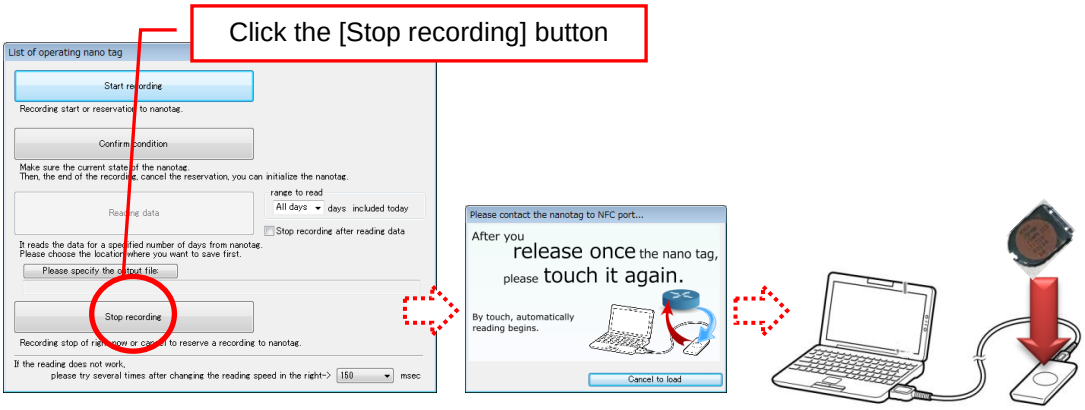
➢ Cancel a reservation

Cancel the reserved recording of "nano tag".

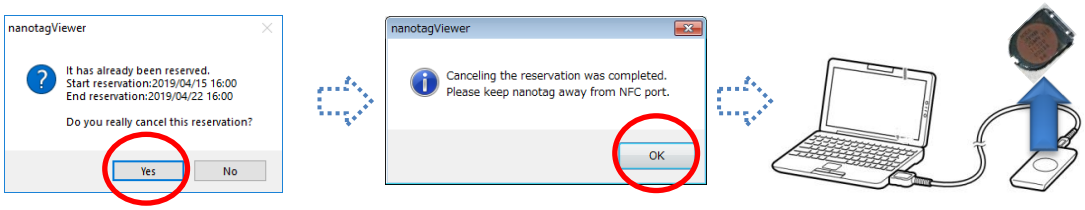
1. After connecting the NFC port to your PC, run "nanotagViewer" and select the [Data] - [List of nano tag operations] menu.

2. Click the [Stop recording] button to move "nano tag" close to the NFC port.

   Move the antenna side (the surface without the text written) of "nano tag" close to the NFC port.



When the screen of "Do you really cancel this reservation?" appears, click the [Yes] button and move "nano tag" close to the NFC port again.
Next, when the screen of "Canceling reservation was completed." appears, release "nano tag" from the NFC port and click the [OK] button.



3.3.4 Read in the data

Read data from "nano tag".
You can read data from "nano tag" during or after recording.

Info

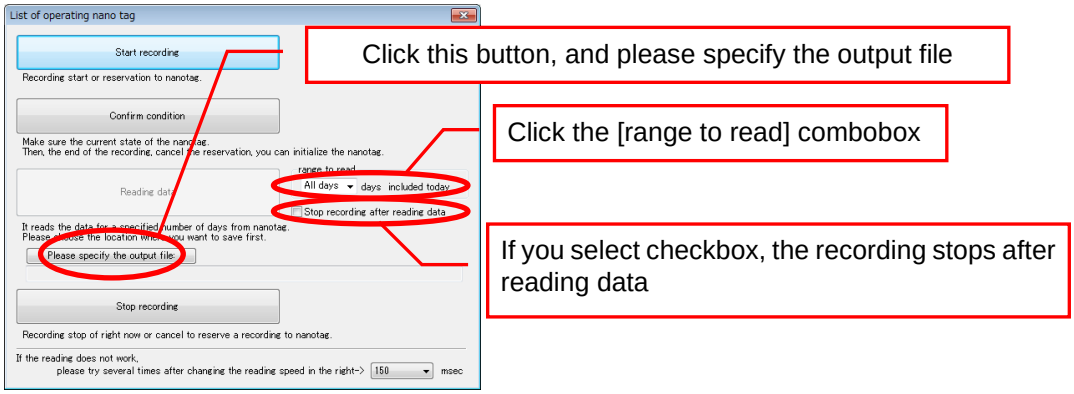
The data in "nano tag" is not erased even if the data is read by "nanotagViewer".

The data in "nano tag" is not erased unless you run the [Initialization] button or another recording.

1. After connecting the NFC port to your PC, run "nanotagViewer" and select the [Data] - [List of nano tag operations] menu.

2. Next, determine which files type to output and how many days to read.

   Click the [Press specify the output file] button and to set the destination for the file.
   Also, select the [range to read].
   "All days" : Read all data in "nano tag".
   If you specify a number of days. : Reads data for specified number of days (including today)



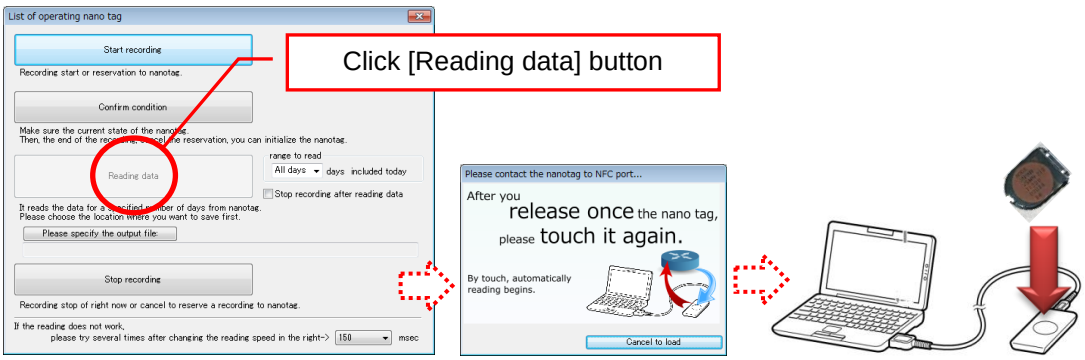
Caution

If you check the [Stop recording after reading data], recording stop automatically after you read data of "nano tag".

Please note that recording stops even if the specified recording time remains.

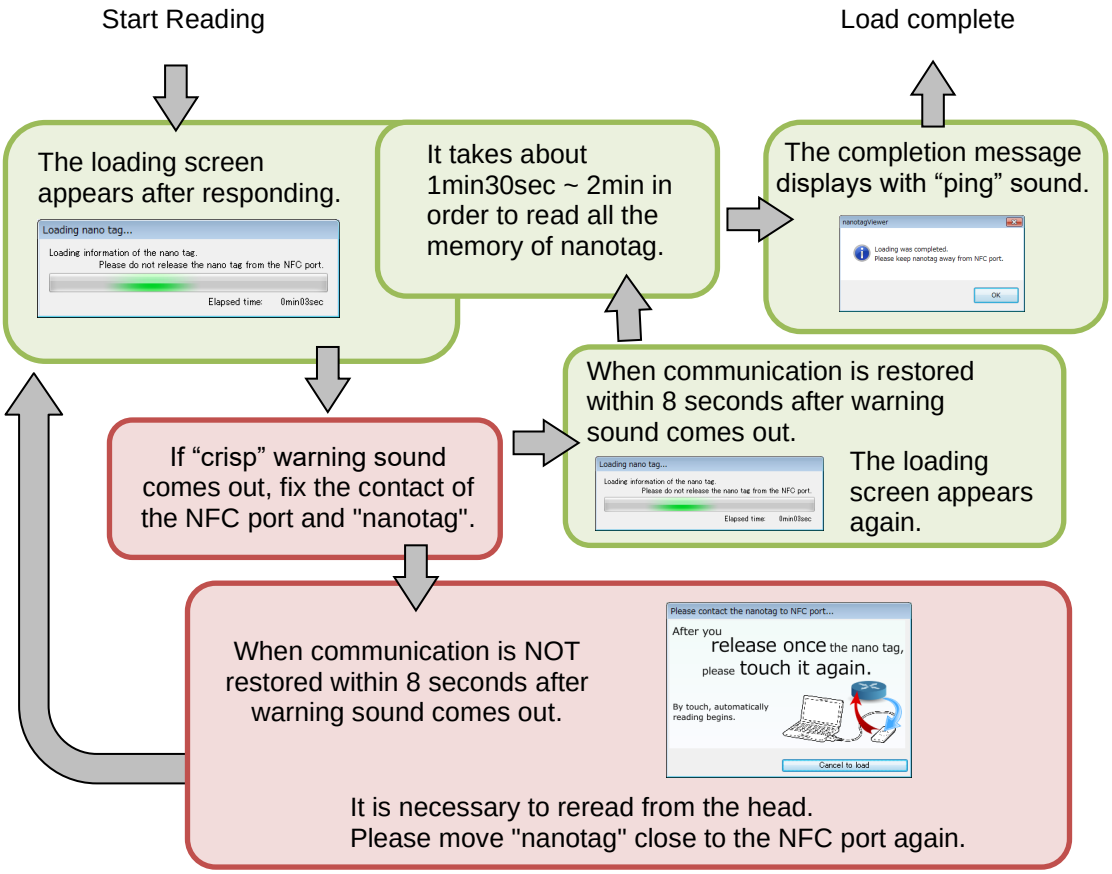
3. Click the [Reading data] button, and move "nano tag" closer to NFC port.

   Move the antenna side (the surface without the text written) of "nano tag" close to the NFC port.

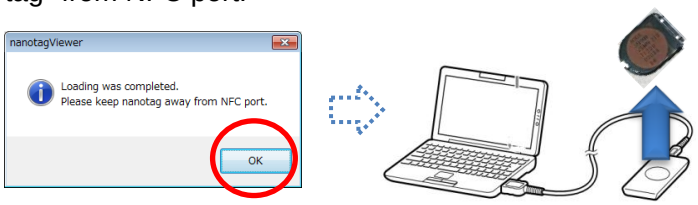


● You can know the status of NFC communication by the beep sound of your PC without looking at the PC screen.

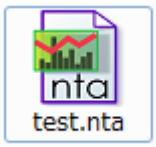
If “crisp” warning sound comes out, fix the “nano tag” and NFC port connection. If communication is restored within 8 seconds after warning sound comes out, continue reading data.



4. When the screen of "Loading was completed." appears with "ping" sound, release "nano tag" from NFC port.



“nanotagViewer” create a data file (extension nta, ntc, or ntr) at the specified destination.



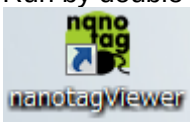
3.4 Analysis of recorded data

This section describes how to open the recorded data file and observe the biological rhythm.

3.4.1 Open the data

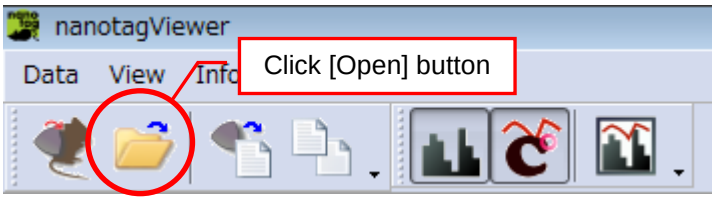
Open already recorded data.

1. Start “nanotagViewer”.
   Run by double-clicking the “nanotagViewer” icon on the desktop.

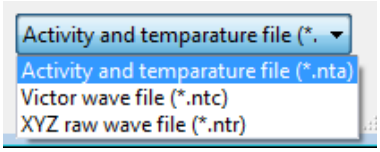


2. Select the [Open] button from toolbar.

   When select the [Open] button from toolbar, the "Open the data file" screen is displayed.



3. Select the "File type" from the screen and open the file.



4. The data is displayed on the screen.

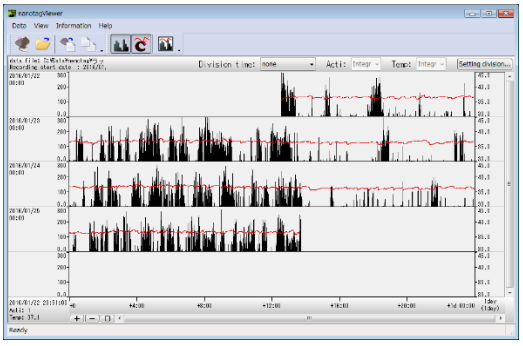
3.4.2 Types of recorded data

■ Activity and temperature file (File extension: .nta）

Activity : Frequency
(Number of times the composition of acceleration in 3 axes exceeded the threshold)
The unit of value is "number of times".

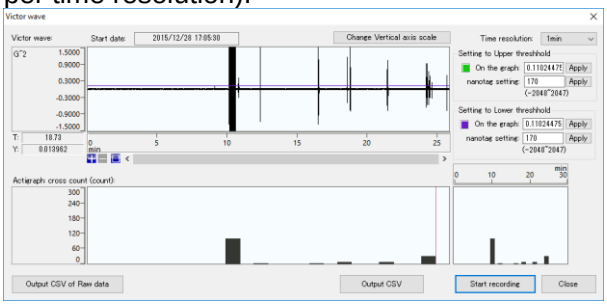
Temperature : Temperature inside the ”nano tag” case.
The unit of value is "degree (Celsius)".

Both data are displayed on the main screen.

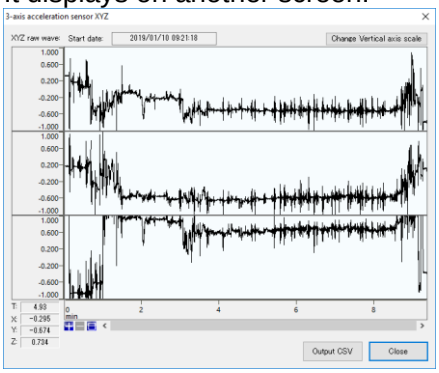


■ Vector wave file [composition of acceleration in 3 axes] (File extension: .ntc)
This is the original waveform to calculate the amount of activity. The unit is no.
It displays on another screen.

You can change the threshold for calculating the amount of activity (number of vibrations per time resolution).



■ XYZ raw wave file (File extension: .ntr)
You can see to the raw data acquired from the acceleration sensor. The unit is "G".
It displays on another screen.



3.4.3 Setting of Display graph

The functions to make it easy to observe the rhythm of activity and temperature are as follows.

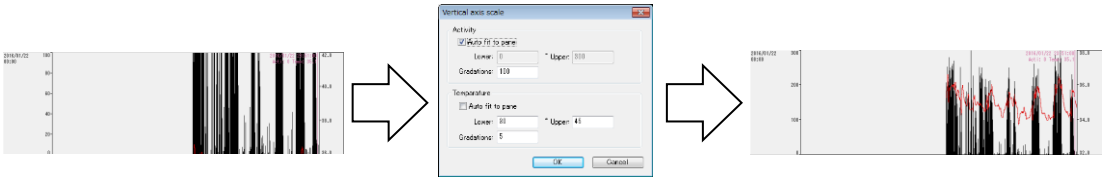
➢ Change the vertical axis scale

The vertical axis of the graph can change in the [Vertical axis scale] menu.

The graph is full.
Not see the temperature data.

Right-click, and select
[Vertical scale] menu

You can make the graph easier to see by changing the vertical scale.



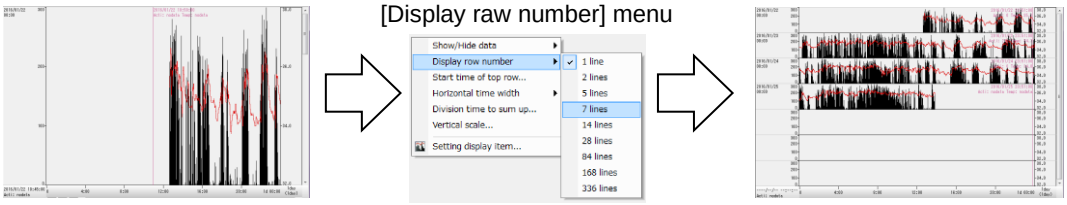
➢ Change the number of lines

The number of graphs can change in the [Display raw number] menu.

1-line

Right-click, and select
[Display raw number] menu

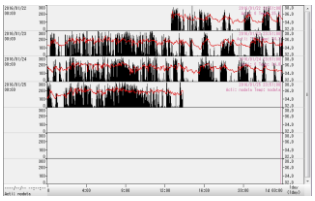
7 lines display



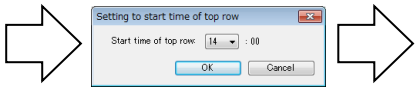
➢ Change the start time of graph

You can change the start time of the graph in the [Start time of top raw] menu.
For example, if the switch from dark phase to light phase is 14 o'clock, setting the graph start time to 14 o'clock makes it easier to observe the rhythm.

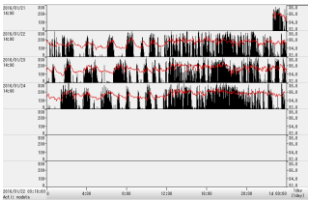
Start at 0 o'clock



Right-click, and select
[Start time of top raw] menu



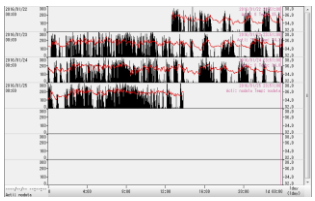
14 o'clock start



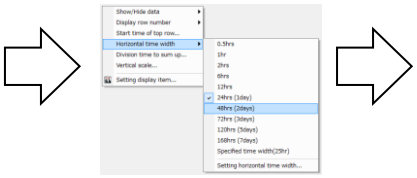
➢ Change the display range of graph

The time width of the horizontal axis of the graph can change in the [Horizontal time width] menu. The time width selected from 11 types prepared in advance or input in 1-hour units.

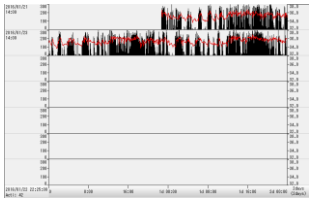
24-hour display



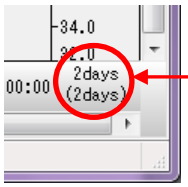
Right-click, and select
[Horizontal time width] menu



48 hours display



The currently set horizontal axis time width can confirm in the display field at the lower right of the screen.

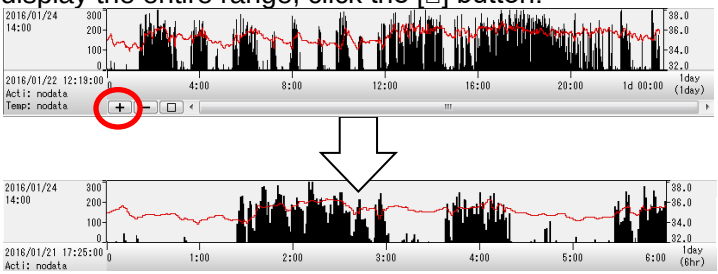


Current horizontal axis display range.
The value in parentheses are the display range when the horizontal axis is enlarged or reduced.

➢ Zoom in or out on the horizontal axis of the graph

You can change the zoom in or zoom out of the graph horizontal axis with the [+] or the [-] button.

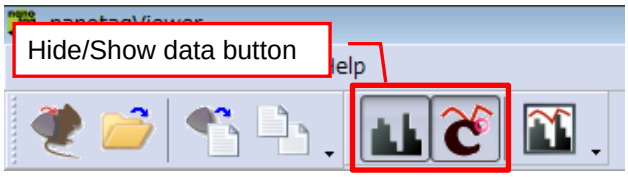
Click the [+] button for zoom in or the [-] button for zoom out.
And, to display the entire range, click the [□] button.



24-hour display

6 hours display

➢ Show or hide waveforms (activity and temperature)



➢ Read the value of the graph

When you move the mouse cursor on the graph, the value of that position is displayed at the lower left of the screen.



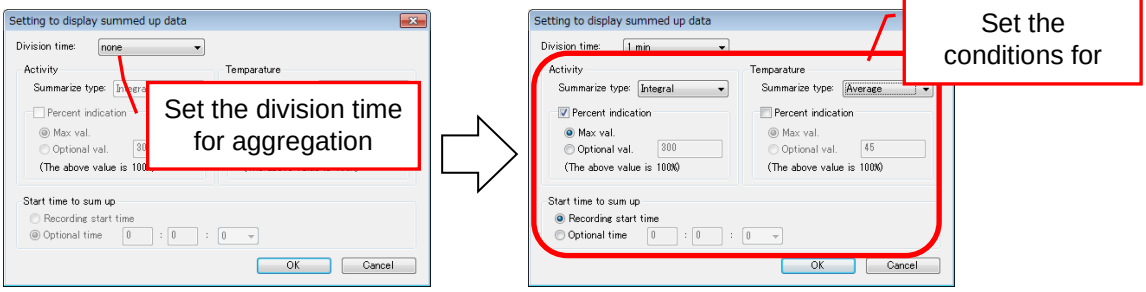
Furthermore, if you left click of mouse on the graph, the value is displayed on the upper right. (Normal mode only, it is not displayed when “Summarize data”.)



3.5 Summarize data

This section describes how to count up the amount of activity and temperature.

1. Click the [Setting division] button.



<Various parameters>

・Summarize type

Integral	Displays the integrated value at each specified time.
Average	Displays the averaged value at each specified time.
Minimum	Displays the minimum value at each specified time.
Maximum	Displays the maximum value at each specified time.

・Percent indication

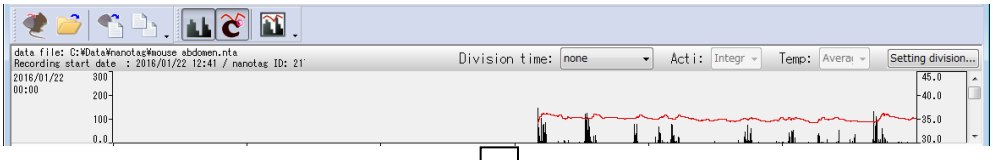
Not use	The values after aggregation are displayed as they are.	
use	Max. val.	After summarizing data, display the percentage to maximum value.
	Optional val.	After summarizing data, display the percentage for the specified value.

・Start time to sum up

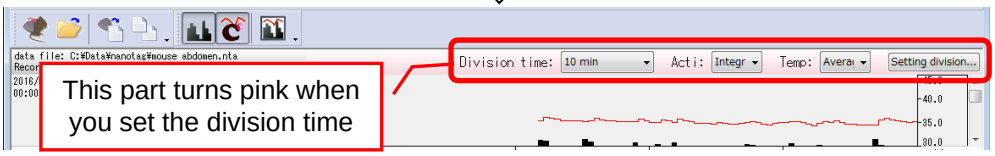
Recording start time	Data aggregation starts from the recording start time.
Optional time	Data aggregation starts from the time you set arbitrarily. Ex.) Using data that started recording at 12:00 on 2017/01/01, If you set the start time to 00:00:00. →Start aggregation from at 00:00:00 on 2017/01/02.

2. After setting the conditions, click the [OK] button.
The graph is displayed under the conditions you specified.

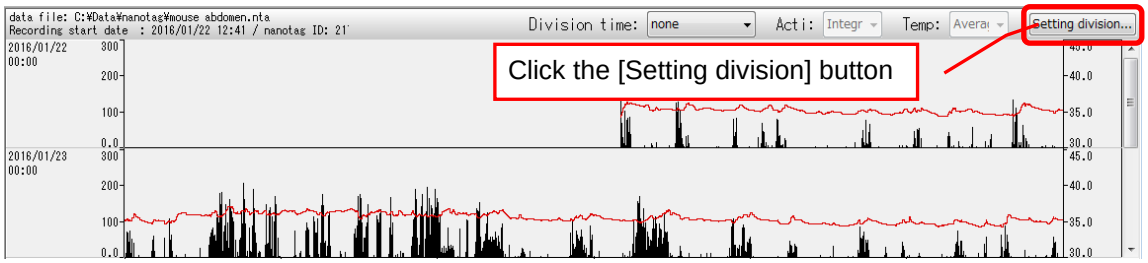
Before



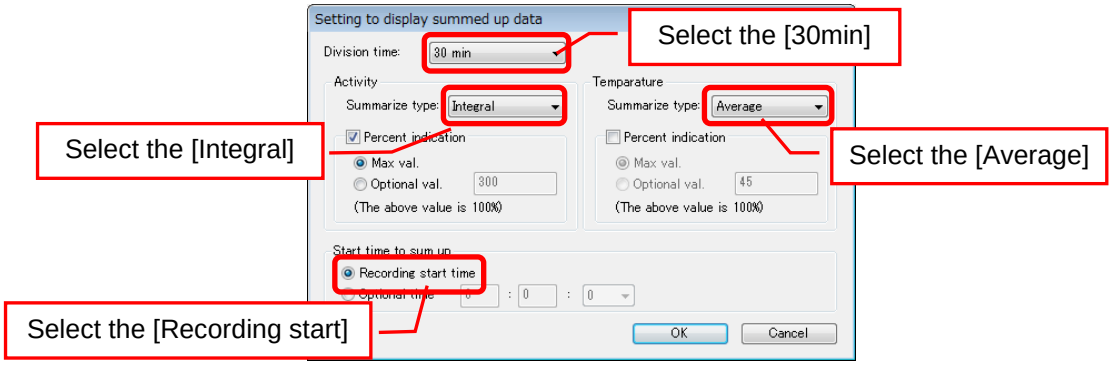
After



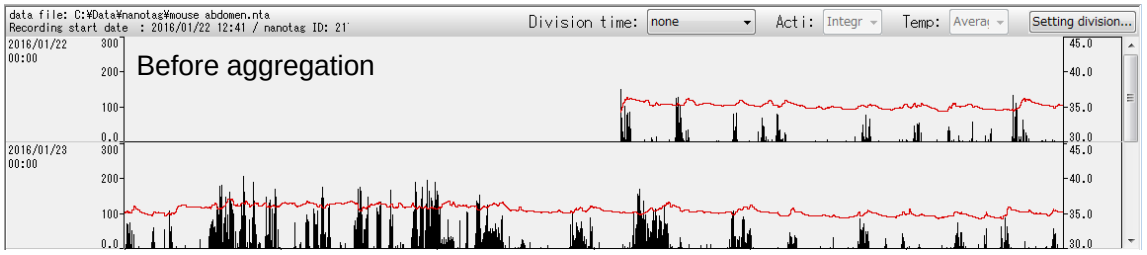
Ex.) If you want to calculate "integral value of activity" and "average value of temperature" from top of data every 30 minutes.

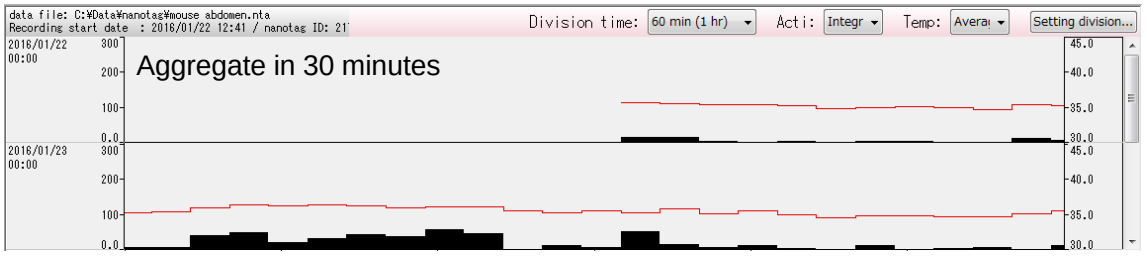


Please set as follows.



The graph changes as follows.

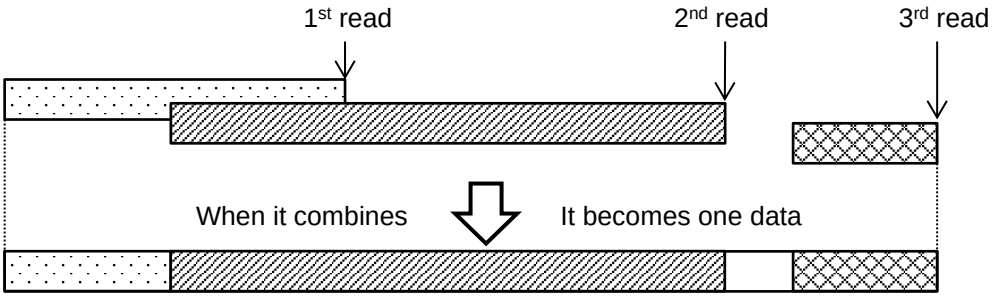




3.6 Combine data

Combine multiple data files into one data.

If you read data from the same "nano tag" every few days, you get multiple files.
"nanotagViewer" can combine these files into a file.



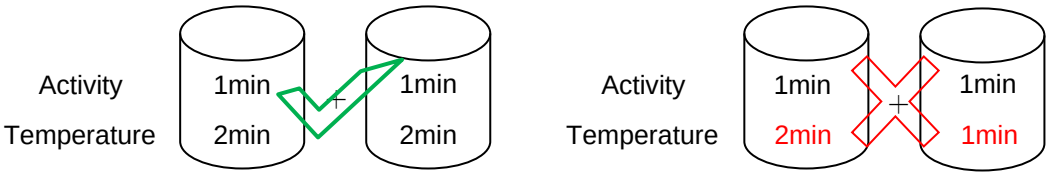
<How to combine>

1. Run "nanotagViewer", and select the [Data]-[Combine recording files] menu.

2. Click the [Add] button at the top of the screen to select multiple files you want to combine.

Caution

You can combine multiple files only if the time resolution of activity and temperature are the same each file.

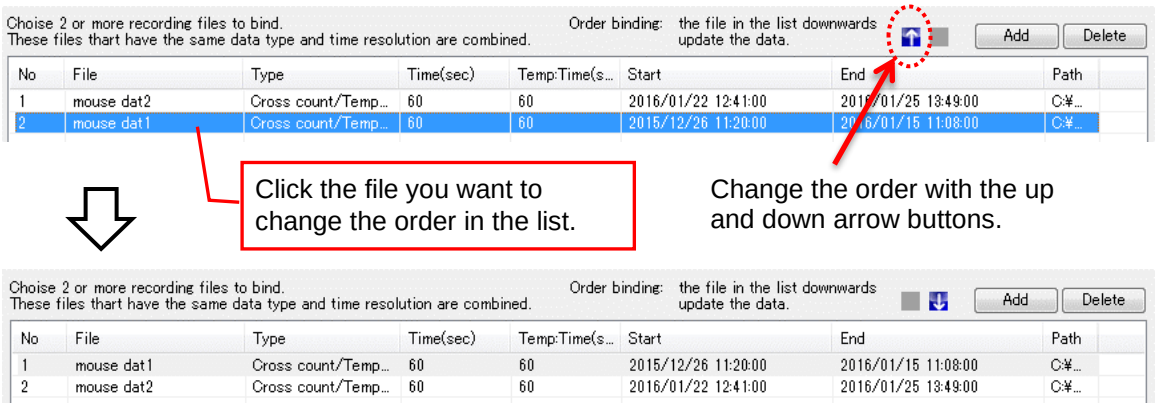


You cannot combine files with different recording conditions as follows.

- Time resolution is different for each file.
- The activity not record.
- The temperature not record.

3. Arrange the order of combine in the list.

"nanotagViewer" combines in order from the top of the list, so keep in mind the order of the list. For more information, please see to the next page [<About combine algorithm>] section.



The order of the files is switched.

4. Select two or more files from the list, click the [OK] button, and then specify the output destination file.

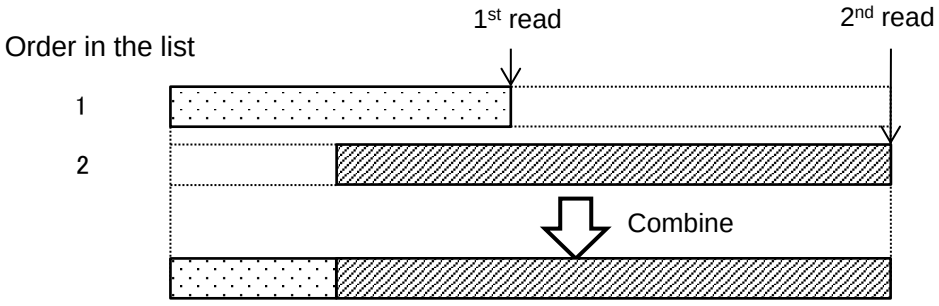
<About combine algorithm>

"nanotagViewer" combines in order from the top of the list.

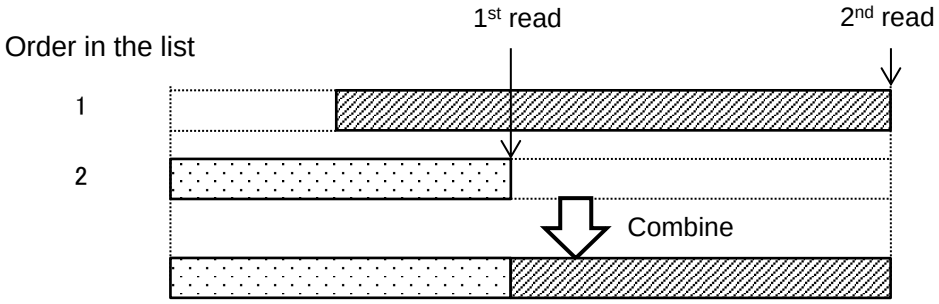
If the data overlap, the data in the lower part of the list has priority.
The data at the top of the list will be overwritten, so be careful of the list order.

1） When the data overlap.

a) If the order of the list is the order of read data from "nano tag"

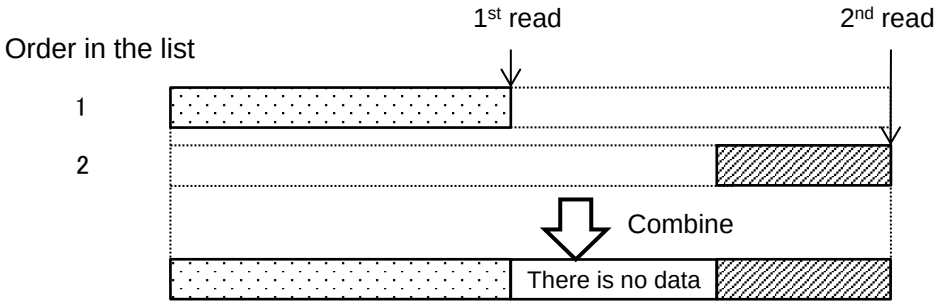


b) If the order of the list is the reverse of a).

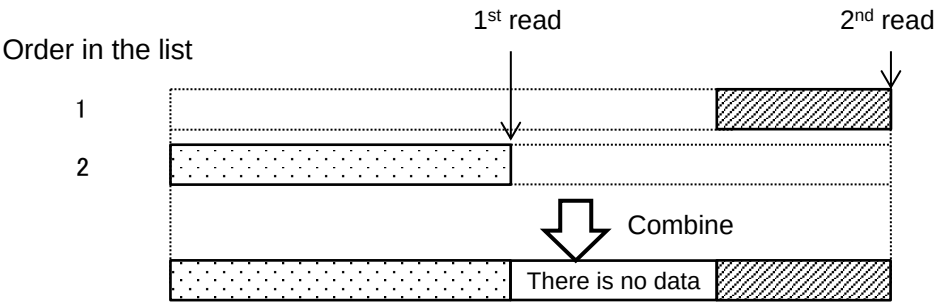


2） When data do not overlap

a) If the order of the list is the order of read data from "nano tag"



b) If the order of the list is the reverse of a).



3.7 Output data

This section describes how to output csv file or print.

3.7.1 Output CSV

There are two ways to output to a CSV file.

- a file (currently display data)
- multiple files at once

The output CSV file can use with spreadsheet software (such as Microsoft Excel).

➢ CSV output of displayed graph

1. Activity and Temperature file (.nta)
   1) Display the data you want to output.
   2) Select the [Data]-[Output CSV file]-[from Active data] menu.
   3) Set the output destination, and click the [OK] button.

Info

When using [Summarize data], "nanotagViewer" output the data summarized as CSV.

2. Vector wave file [composition of acceleration in 3 axes] (.ntc)
   ○ Data of Vector wave and Actigraph
   1) Display the data you want to output.
   2) Click the [Output CSV] button at the bottom right of the screen.
   3) Set the output destination, and click the [OK] button.

   ○ Raw data of Vector wave [composition of acceleration in 3 axes]
   1) Display the data you want to output.
   2) Click the [Output CSV of Raw data] button at the bottom left of the screen.
   3) Set the output destination, and click the [OK] button.

3. XYZ raw wave file (.ntr)
   1) Display the data you want to output.
   2) Click the [Output CSV] button at the bottom right of the screen.
   3) Set the output destination, and click the [OK] button.

➢ CSV output multiple files at once

1. Select the [Data]-[Output CSV file]-[from Multiple files] menu.

2. Select multiple files (Up to 100 files)

3. Set the output destination, and click the [OK] button.

Info

When using [Summarize data] in "Activity and Temperature file (.nta)", "nanotagViewer" output the data summarized as CSV. In this case, it outputs with the file name of "original file name_Xmin.CSV".

➢ Contents of CSV

1. Activity (cross count) and temperature (.nta)

○ Normal output (Do not summarize)

Time resolution

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Time resolution	Activity(Cross count)	60	sec	Temparature	60	sec								
2	Recording conditions	nanotag ID:2170000073/Upper threshhold:140/Lower threshhold:140/Start date_Activity:2016/01/22 12:33:00/Temparature range (degC):30.0~45.8/Start date_Temparat													
3															
4	Date	Cross count	Temparature												
5	2016/1/22 12:33	170	35.75												
6	2016/1/22 12:34	313	35.6875												
7	2016/1/22 12:35	279	35.6875												
8	2016/1/22 12:36	193	35.75												
9	2016/1/22 12:37	221	35.875												
10	2016/1/22 12:38	191	36.0625												
11	2016/1/22 12:39	278	36.0625												

Setting conditions when recording

Date:
Please display the time in seconds on Microsoft Excel.
Ex.) “yyyy/mm/dd hh:mm:ss”

○ Summarize data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Time resolution	Activity(Cross count)	60	sec	Temparature	60	sec							
2	Recording conditions	nanotag ID:2170000073/Upper threshhold:140/Lower threshhold:140/Start date_Activity:2016/01/22 12:33:00/Temparature range (degC):30.0~45.8/Start date_Tempar												
3	Division time to sum up	5	min											
4	Date	Cross count(Integral)	SE	Temparature(Average)	SE									
5	2016/1/23 0:00	104		36.75	0									
6	2016/1/23 0:05	12		36.837502	0.025									
7	2016/1/23 0:10	5		36.9375	0									
8	2016/1/23 0:15	47		37.025002	0.015309									

Division time to sum up

Result after aggregation

Standard Error (SE):
The standard error is only output if the aggregation conditions are set to "average" and "non-percentage".

2. Vector data [composition of acceleration in 3 axes] (.ntc)

○ Data of Vector wave and Actigraph

Data resolution
[fixed at 25Hz]

Time resolution

Setting conditions of threshold

	A	B	C	D	E	F	G	H	I
1	Time resolution	Victor wave	25	Hz	Summarization range	60	sec		
2	Conditions	Upper threshhold_V	0.110245	Upper threshhold_Se	170	Lower threshhold	0.110245	Lower threshh	170
3	nanotag ID	0							
4									
5	Date	Victor wave(G^2)	Victor wave_spline(G^2)	Cross count					
6	2015/12/28 17:05:30.000	0	0	0					
7	2015/12/28 17:05:30.040	-0.019775	-0.019051						
8	2015/12/28 17:05:30.080	-0.020573	-0.021273						
9	2015/12/28 17:05:30.120	-0.014423	-0.014494						
10	2015/12/28 17:05:30.160	-0.01522	-0.017925						
291	2015/12/28 17:05:41.400	0.009025	0.004541						
292	2015/12/28 17:05:41.440	0.001216	-0.001163						
293	2015/12/28 17:05:41.480	No data	-0.012889						
294	2015/12/28 17:05:41.520	No data	-0.008129						
295	2015/12/28 17:05:41.560	No data	-0.005611						
296	2015/12/28 17:05:41.600	No data	-0.009315						
297	2015/12/28 17:05:41.640	No data	0.00624						
298	2015/12/28 17:05:41.680	No data	0.010174						
299	2015/12/28 17:05:41.720	No data	0.000726						
300	2015/12/28 17:05:41.760	No data	-0.005317						
301	2015/12/28 17:05:41.800	No data	-0.003526						
302	2015/12/28 17:05:41.840	No data	-0.007515						
303	2015/12/28 17:05:41.880	No data	0.002316						
304	2015/12/28 17:05:41.920	No data	0.002582						
305	2015/12/28 17:05:41.960	No data	-0.002888						
306	2015/12/28 17:05:42.000	-0.009808	-0.009808						
307	2015/12/28 17:05:42.040	0.013092	0.013677						

Activity (cross count)

1) Vector data (composition of acceleration in 3 axes) [Unit: G^2]

2) Spline-interpolate 1) every 6 seconds, also applies to "no data". [Unit: G^2]

Date:
Please display the time in milliseconds on Excel.
Ex.) “yyyy/mm/dd hh:mm:ss.000”

○ Raw data of composition of acceleration in 3 axes

Data resolution [fixed at 25Hz]

	A	B	C	D	E	F	G
1	Time resolution	Victor wave_Raw(G)		25 Hz	Summarization range	60	sec
2							
3	nanotag ID	0					
4							
5	Date	Raw(G)	Raw_spline(G)				
6	2015/12/28 17:05:30.000	0.849609	0.849609				
7	2015/12/28 17:05:30.040	0.837891	0.838391				
8	2015/12/28 17:05:30.080	0.835938	0.835785				
9	2015/12/28 17:05:30.120	0.837891	0.83822				
10	2015/12/28 17:05:30.160	0.835938	0.834871				

1) Raw data of composition of acceleration in 3 axes : $\sqrt{X^2 + Y^2 + Z^2}$

2) Spline-interpolate 1) every 6 seconds, also applies to "no data". [Unit: G^2]

Date:
Please display the time in milliseconds on Excel.
Ex.) “yyyy/mm/dd hh:mm:ss.000”

3. XYZ data (.ntr)

Data resolution [fixed at 25Hz]

Each value of X, Y, Z [Unit: G]

	A	B	C	D
1	Time resolution	XYZ	25	Hz
2	nanotag ID	0		
3				
4	Date	X(G)	Y(G)	Z(G)
5	2015/12/28 17:47:30.000	0.085938	0.029297	-0.84375
6	2015/12/28 17:47:30.040	0.083984	0.027344	-0.841797
7	2015/12/28 17:47:30.080	0.085938	0.029297	-0.837891
8	2015/12/28 17:47:30.120	0.076172	0.023438	-0.835938
9	2015/12/28 17:47:30.160	0.089844	0.029297	-0.837891
10	2015/12/28 17:47:30.200	0.080078	0.023438	-0.835938

Date:
Please display the time in milliseconds on Excel.
Ex.) “yyyy/mm/dd hh:mm:ss.000”

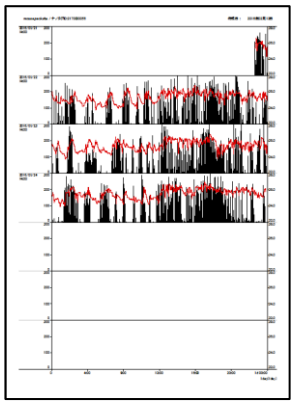
3.7.2 Print the graph

Prints the graph currently displayed on the screen.

1. Display the graph you want to print.
   If you want to change the display format of the graph to be printed, please see to “3.4.3 Setting of Display graph (p38)” section.

2. Select the [Data]-[Print graph] menu.

3. Set up the printer and print.
   For print preview, select the [Data]-[Print preview graph] menu.



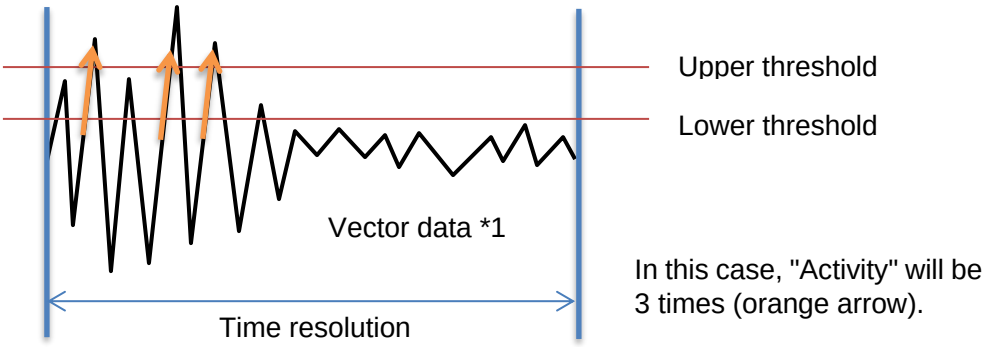
4 Supplement

4.1 Calculation method of Activity (cross counts)

This section describes how to count up of activity.

“nano tag” automatically converts the acceleration sensor values (X, Y, Z values) into vector data (composition of acceleration in 3 axes).

"nanotagViewer" counts the number of times the threshold is crossed from the bottom to the top within the specified time resolution from this vector data.
And this count value is Activity (cross counts), and the unit is number of times.



*1: composition of acceleration in 3 axes

The recommended values examined by us for the threshold are as follows.

	Mouse		Rat	
	neck	abdomen	neck	abdomen
Threshold in 1 min steps (number of times)	170	150	170	140

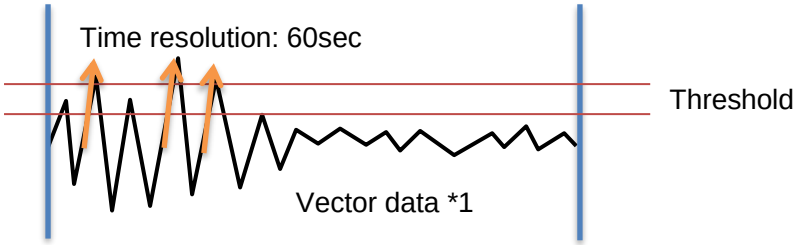
4.2 Calculation range of activity and temperature

This section describes the CSV format of activity and temperature data (.nta).

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Time resolution	Activity(Cross count)	60 sec		Temparature	60 sec									
2	Recording conditions	nanotag ID:2170000073/Upper threshhold:140/Lower threshhold:140/Start date_Activity:2016/01/22 12:33:00/Temparature range (degC):30.0~45.8/Start date_Tempar													
3															
4	Date	Cross count	Temparature												
5	2016/1/22 12:33	170	35.75												
6	2016/1/22 12:34	313	35.6875												
7	2016/1/22 12:35	3	35.6875												
8	2016/1/22 12:36	193	35.75												
9	2016/1/22 12:37	221	35.875												

Ex.) The meaning of the data value at this time is as follows.

○ Activity (cross counts)



2016/01/22 12:35:00

2016/01/22 12:35:59
The number of times up to this point is output as "Activity".

○ Temperature

2016/01/22 12:35:00
"Temperature" at this point is output.

Revision history

Version	Date	Content
1.2.0.1906	Jun 30, 2019	First edition
1.2.1.1908	August 30, 2019	Added model VSMN210
1.2.1.2007	July 20, 2020	Some cautions were added to the procedure to start recording in the demo version.
1.2.4.2509	Sep 25, 2025	Changed PC specifications and FeliCa reader device configuration method